HTC Core: Integrated Procurement & Financial Tracking System

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CHAPTER 1

THE PROBLEM AND ITS BACKGROUND

Introduction

Heindrich Trading Corporation (HTC) is a Makati-based bulk commodity trader and a vital link in the alcohol industry. Despite its lean size, HTC secures massive molasses supply agreements, managing complex land and sea logistics directly from industrial sugar mills like BUSCO to beverage giants such as San Miguel, Tanduay, and Emperador. In 2026 alone, HTC handled 89,190.739 metric tons (MT) in deliveries (Heindrich Trading Corporation, Appendix E, 2026). However, operating on tight cash-flow margins with massive cargo volumes creates severe financial exposure. To fund upfront procurement and settle 50% logistics deposits, HTC relies heavily on short-term bank loans. Because client payments frequently stretch beyond standard 30-day windows, the company faces compounding bank interest that eats directly into profit margins.

The gravity of this exposure is highlighted in their company books as shown in Appendix B: despite an accepted 1% volume tolerance for transit tank fluctuations, HTC still incurred a staggering net loss of ₱2,343,480.55 (Heindrich Trading Corporation, Appendix A, 2026). This loss stems directly from three unmanaged areas: transit variances, operational complexities, and overdue loans. Specifically, client-specific leaks show losses scaling upwards from BDI (₱320,267.85) and Progreen (₱327,338.40), spiking through GSMI (₱439,866.0) and Emperador (₱511,759.50), and peaking heavily with ADI at ₱744,248.80 (Heindrich Trading Corporation, Appendix E, 2026).

Presently, these variables including cash vouchers, bank loans, and barge logistics are managed via a fragmented, manual workflow of physical index cards and a single local Excel file. This manual

dependency creates a dangerous blind spot: staff cannot manually compute daily compounding interest or cross-route volumetric shrinkage fast enough to catch margin erosion before it occurs. To mitigate these risks, HTC needs to transition to a centralized digital framework. By bypassing expensive physical hardware sensors, this study implements an intelligent, pure-software solution. Developing a dedicated platform featuring dynamic interest accrual tracking, asynchronous background transit volume calculations, and strict user permission boundaries will eliminate manual leaks, optimize capital loan lifecycles, and secure HTC's financial bottom line.

Statement of the Problem

Heindrich Trading Co. currently manages its complex bulk procurement, sales logging, and logistics tracking through manual, spreadsheet-reliant legacy workflows. This operational setup leads to several core problems:

1. Centralized Data Fragmentation and Cross-Generational Operational Friction: Because HTC's lean team operates across contrasting generational methodologies ranging from localized spreadsheet consolidation by new hires to physical index-card tracking by senior personnel, transactional updates remain heavily isolated. This data fragmentation creates severe blind spots where real-time operational status updates cannot be shared concurrently across departments.
2. Cognitive Overstimulation and Risk Exposure in Manual Cash-Flow Reconciliation: The financial workflow demands a 50% upfront downpayment to logistics providers before receiving client settlements, requiring staff to manually cross-reference check payouts with complex multi-load logistics invoices. Managing this multi-tiered matching process through raw, number-heavy sheets creates high cognitive overstimulation for senior users, increasing the risk of delayed reconciliation and unmonitored profit margin erosion.
3. Passive Volumetric Loss Monitoring and Lack of Centralized Verification: Due to environmental factors and moisture drops during bulk commodity transport, the volumes loaded at sugar mills

fluctuate before final delivery. Without a single-click verification interface, these critical transit discrepancies are reviewed retrospectively on paper, making it difficult for the current personnel layout to efficiently monitor shrinkage and apply immediate financial adjustments.

We identified these operational challenges by conducting an extensive one-on-one interview with an Operations Staff member; please refer to Appendix A for the complete breakdown of the interview transcript

Project Objectives

General Objective

To develop HTC Core, a secure and easy-to-use web system that replaces Heindrich Trading Corporation's paper and spreadsheet workflows with one centralized platform for procurement, finance, and logistics.

Specific Objectives

1. To centralize fragmented records into a unified dashboard with consolidated data visualization and real-time status monitoring
2. To automate the management of bank loans, cheque transactions, and 50% upfront logistics deposits, including interest computation and repayment notifications.
3. To enforce shipment volume variance monitoring against the 1% tolerance threshold and maintain an immutable audit trail of financial activities to strengthen fraud prevention.

Significance of the Study

The development and implementation of this cloud-based Management Information System are driven by a primary mandate: to modernize, secure, and streamline the core operational ecosystem of

Heindrich Trading Co., specifically engineered to optimize business oversight and bring maximum convenience to its management and employees. By transforming fragmented, manual tracking routines into a unified, high-performance cloud environment, this study delivers profound strategic and practical value to its stakeholders:

Ms. Marinic Perez: This project is structurally dedicated to providing total convenience, structural clarity, and administrative ease to corporate management. Recognizing that executive decision-making requires instant access to accurate data, this system eliminates the daily friction of auditing manual spreadsheets and chasing disjointed reports. It presents our stakeholders with a centralized, elegant administrative dashboard that consolidates complex logistics, procurement, and financial data into clear, real-time business reports. Furthermore, because the enterprise deals in life-changing amounts of money via physical cheques, the system specifically provides ownership with an Immutable System Audit Trail. This unalterable security layer allows management to monitor high-value check movements and data modifications with absolute security and fraud protection, granting total peace of mind over the company's trading lifecycle.

Operations and Invoicing Staff: The system significantly eases the daily operational burden on logistics and frontline documentation staff by replacing tedious manual coordination with structured automated workflows. Built-in database constraints prevent entry errors while providing a clean, high-contrast digital interface that simplifies complex data-entry workflows. Most importantly, rather than forcing personnel to calculate volumetric transit shrinkage manually on paper, the system utilizes asynchronous background scripts to automatically run volume variance calculations, instantly triggering visual alarms if cargo drops exceed standard tolerances. This guarantees immediate operational consistency and lowers execution risks.

Finance Personnel: This study introduces native financial intelligence that removes the heavy lifting from the company's fiscal workflows. Crucially, the system frees senior finance staff from manually tracking compounding bank loan interest on physical index cards. By deploying a dynamic interest accrual engine, the platform provides automated payment-to-expense matching scripts that seamlessly tie sales invoices to cash vouchers, active bank loans, and 50% upfront logistics deposits. This protects the ledger from human arithmetic errors and accelerates settlement timelines by visually alerting staff precisely when to close active loans to prevent margin erosion.

The Industry and Small-Scale Commodity Distributors: This study establishes an empirical precedent proving that independent trading companies can attain high-level enterprise resource planning (ERP) capabilities without taking on the capital-intensive infrastructure dependencies (e.g., dedicated hardware arrays, private blockchain architectures) heavily criticized in modern literature. It offers a low-cost, software-driven blueprint for organizational modernization.

Future Researchers and Academics: This research contributes a reliable academic framework for the design and deployment of secure, high-concurrency cloud systems utilizing a decoupled multi-tier stack (NGINX, Gunicorn, Django, Celery/Redis, and PostgreSQL managed cloud infrastructure). It serves as a thorough reference for future scholars specializing in secure application engineering and iterative waterfall development methodologies within localized commodity trading markets.

Scope and Limitations

Scope

The project focuses on building a dedicated web platform to manage the core local trading, logistics, and internal reconciliation operations of Heindrich Trading Co. The system scope is defined by the following features:

1. Core Modules: Contract and Purchase Order tracking, multi-variable logistics ledger (barge and tracking service records), dynamic input for transit quantity variances (delivered vs. received volume), and dual-invoice cross-referencing (supplier invoice numbers linked to internal Cash Voucher and Sales Invoice numbers).

2. Automation Logic: Real-time software alerts driven by database constraints to flag contract fulfillment progress, missing expense associations, and significant volume discrepancies exceeding standard tolerance thresholds.

3. Access and Control Security: A secure login framework with multi-role permissions that splits views and capabilities between administrative users and standard employees.

4. Deployment Architecture: A cloud-hosted web system that runs efficiently across modern desktop web browsers, ensuring remote accessibility and collaborative data updating without local server maintenance.

Limitations

To maintain project focus and stay within budget, the system operates under the following boundaries:

1. Hardware Exclusions: The platform uses pure software and database logic to log transaction records and track logistical costs. It does not integrate with external hardware tracking arrays, such as RFID chips, barcode scanners, or automated GPS tracking units hosted on physical barges.

2. Financial and Accounting Exclusions: The system focuses strictly on operational procurement, sales logging, and transactional payment-to-expense matching. It does not provide full corporate

accounting capabilities, general ledger generation, balance sheet balancing, or automated tax localization modules.

3. Financial Confidentiality and Structural Exclusions: Heindrich Trading Co. has reserved the right to keep overall corporate financial statements strictly confidential. Consequently, while the platform tracks transactional interest accruals, cheque disbursements, and shipment-level cost adjustments, it explicitly excludes macro profit-and-loss auditing, multi-currency ledgers, tax localization modules, and automated tax filing systems.

4. E-Commerce Boundaries: The interface is built strictly for internal business administration, management tracking, and back-office billing log synchronization. It does not include a public-facing customer storefront, broker interaction portals, or online customer payment gateways.

5. SRA Regulatory Automation Limits: While the platform stores sourcing data types (local vs. imported molasses metrics), it will not dynamically automate or enforce complex Sugar Regulatory Administration compliance rules (such as the 3:1 local-to-import rule or customs clearance document tracking) within the system architecture. These remain external compliance workflows.

6. Tank Fluctuations: Environmental volume shifts are a fixed physical constraint.
   We absorb the 1% transit tolerance as per the company's protocol.

Definition of Terms

The following technical and operational terms are defined according to their specific contextual application within the HTC Core platform and the operational environment of Heindrich Trading Corporation. These terms reflect all architectural components, cryptographic audit subsystems, and financial workflow

controls developed for the study:

ACID Compliance – ACID Transaction Compliance refers to the mathematical and database-level guarantees enforced by our PostgreSQL 16 relational database engine and Django's transaction.atomic() blocks to ensure that high-value financial records, credit allocations, and bulk molasses delivery operations never enter a corrupted, incomplete, or phantom state.

Blockchain Ledger – A cryptographic security subsystem within the platform that links sequential operational transactions, purchase order milestones, and credit facility modifications into an immutable SHA-256 block hash chain, providing verifiable, tamper-evident audit proofs for executive management.

Bulk Commodity Trading – The commercial buying and selling of large quantities of raw, unpackaged goods (specifically molasses in this study) requiring specialized multi-modal transport, weighbridge calibration tracking, and high-volume contract management.

Business Rules – Programmatic constraints hard-coded into the application logic to automate and enforce company policies, such as blocking duplicate entries, restricting destructive database updates, or triggering alerts when transit volume discrepancies exceed accepted tolerance limits.

Capital Loan Tracking – The digital logging and chronological monitoring of short-term commercial bank financing lines and associated physical paper cheques secured by the enterprise to fund upfront procurement and logistics operations.

Cash Voucher Number – A unique internal alphanumeric identifier generated by the system to authorize, track, and balance cash disbursements distributed upfront to logistics providers, haulers, and sugar mills.

CHAI Record – A quality assurance test record capturing laboratory Brix percentages, total reducing sugars (TRS), and density specifications for bulk molasses shipments prior to executive payment sign-off.

Compensating Correction Pattern – An enterprise accounting and software design pattern wherein erroneous data entries are never directly overwritten or deleted via SQL; rather, the original record is permanently retained as an indelible historical artifact while a linked, offsetting transaction is created with mandatory justification and dual-authorization sign-off.

Creditable Expanded Withholding Tax (EWT) – A localized Philippine tax deduction (1% or 2%) applied programmatically during sales invoicing and supplier payment reconciliations in compliance with Bureau of Internal Revenue (BIR) regulations.

Dynamic Interest Accrual Engine – A specialized accounting calculation module that dynamically computes daily interest liabilities on active commercial bank loans using exact day-count conventions and arbitrary-precision decimal mathematics.

Financial Operations & Cheque Ledger – The unified core financial module of HTC Core that consolidates commercial credit facilities, interest aging clocks, physical cheque disbursements, 50% upfront logistics deposits, and loan settlement lifecycles into a single operational ledger.

Gunicorn (Green Unicorn) – A Python WSGI HTTP application server that translates web requests forwarded from NGINX into dynamic application worker processes executing Django business logic.

High-Legibility Mode – A dedicated user interface accessibility mode engineered to comply with Web Content Accessibility Guidelines Level AA standards by dynamically elevating contrast ratios above 7:1, increasing typography sizing to 1.15rem, expanding line height to 1.6, and eliminating color-only information cues for elderly or field-based personnel.

Immutable System Audit Trail – An unalterable, append-only security ledger within the database architecture that permanently timestamps and records user identities, IP addresses, pre-change/post-change state snapshots, and financial modifications to prevent high-value fraud.

In-Memory Data Structure Store – A high-performance caching framework implemented via Redis that acts as an in-memory data store and message broker to coordinate and buffer background worker queues.

Invoicing Staff – The authenticated user role assigned to billing personnel responsible for inputting outward-facing billing logs, generating BIR-compliant delivery sales invoices, and cross-referencing customer receivables.

ISO/IEC 25010 – An international standard defining a comprehensive quality model used to evaluate systems and software products across functional suitability, performance efficiency, usability, reliability, security, and maintainability.

Iterative Waterfall Model – A modified version of the classical waterfall Software Development Life Cycle (SDLC) that incorporates targeted feedback loops across development sprints, allowing developers to revisit requirements, design, or implementation phases upon stakeholder evaluations.

Legacy Workflow – The traditional, manual, and paper-reliant business practices historically utilized by Heindrich Trading Co., which heavily depended on physical index cards and decentralized, unversioned Microsoft Excel spreadsheets.

Molasses Release Order (MRO) – An official permit and commodity allocation document issued by sugar mills authorizing the withdrawal, physical loading, and transport of specified metric tons of molasses.

Multi-Tier Role Permissions – A software security framework that restricts or grants access to specific views, endpoints, and database models based on an authenticated user's organizational role, enforced at both the HTTP routing and interface layers.

NGINX – A high-performance reverse proxy and web server responsible for handling SSL/TLS cryptographic termination, security header enforcement, rate-limiting, and static file delivery.

Operations Staff – The authenticated user role designated for personnel overseeing field logistics, contract encoding, mill dispatches, barge movements, and transit volume variance monitoring.

Profit Margin Erosion – The unmonitored shrinkage of operational trading margins caused by uncaptured transit losses, unrecorded loan interest accruals, manual data-entry discrepancies, or uncollected supplier deductions.

Purchase Order – A core digital transactional cluster generated upon executive approval, capturing agreed pricing, contracted metric tonnages, sourcing mills, hauling allocations, and terms for a bulk commodity trade.

Redis – An open-source in-memory data structure store utilized as a fast distributed message broker to queue and coordinate tasks executed by Celery asynchronous background workers.

Role-Based Access Control (RBAC) – A formal security architecture that restricts system operations and resource access to authorized users based on predefined organizational privileges.

Sales Invoice Number – The unique, system-generated digital identifier mapped to an outward transaction billing log to track client accounts receivable and settlement balances.

Settled Loan Archival Workflow – A multi-stage lifecycle management workflow that transitions fully reconciled financing facilities from active operational tracking into a closed status, and subsequently into an indexed historical archive vault (is_archived = True) protected by dual-step confirmation dialogs.

Single-Click Execution Model – An ergonomic user interface design methodology that abstracts complex underlying database transformations behind intuitive, high-contrast action triggers to minimize cognitive overhead for non-technical users.

Transit Quantity Variance – The calculated numerical difference between the volume of molasses loaded at

the sugar mill versus the volume confirmed upon delivery at the customer facility, primarily caused by physical transit factors such as evaporation and temperature shifts.

Volume Variance Tolerance Threshold – The exact mathematical boundary (±1.00%) applied to the calculated difference between dispatched and received commodity metric tonnage, beyond which the system programmatically flags a delivery as disputed and halts final invoicing.

WCAG - (Web Content Accessibility Guidelines) is an international W3C standard that provides technical recommendations to make web content accessible to people with disabilities, organized under four principles: Perceivable, Operable, Understandable, and Robust (POUR)

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter establishes the technical, operational, and empirical foundations for the proposed management information system at Heindrich Trading Co. Centralizing and securing data pipelines is now imperative to eliminate fragmented legacy environments and sustain organizational throughput. Specifically, this review explores four unified themes:

1. Web-Based and Cloud-Based Supply Chain Architectures, which outline the structural blueprints for distributed system deployment;
2. Management Control Frameworks and System Integrity Validations, defining programmatic control layers and multi-tier security boundaries.
3. Telemetric Networks, Distributed Ledgers, and Real-Time Asset Tracking, evaluating high-performance tracking and database logic paradigms.
4. Systematization and Meta-Analyses of Digital Transformation Trends, providing macro-level validation for systemic digitization.

By evaluating these models, this review isolates the practical constraints of capital-intensive frameworks, establishing an empirical justification for a software-driven, cost-effective web platform designed to optimize multi-variable contract tracking, contract-to-expense matching, and collaborative transaction updating.

Web-Based and Cloud-Based Supply Chain Architectures

Transitioning from physical reporting to scalable web and cloud infrastructure is critical for fluid workflows. Yu and Zhou (2020) demonstrated that multi-tier supply chain platforms using distributed relational databases effectively bridge information silos among suppliers, manufacturers, and consumers. Node optimization and network synchronization protocols further accelerate data processing speeds across these distributed database clusters (Wang, 2021). Consequently, migrating organizational data to centralized cloud repositories significantly optimizes cross-departmental coordination and overall operational performance (Dong & Salwana, 2022). While these architectures provide robust enterprise blueprints, they focus on macro-level operations with extensive capital backing, omitting localized workflows for independent commodity distributors who must handle rapid sourcing shifts due to volatile supplier inventories. The software platform for Heindrich Trading Co. addresses this gap by adapting multi-tier concepts into a lightweight, collaborative web portal, replacing a manual, single-user legacy setup with real-time cloud data entry and contract tracking.

Management Control Frameworks and System Integrity Validations

To protect these pipelines, transparent data monitoring configurations, immutable transaction logs, and rigid role validations mitigate fraud and entry blind spots (Leong et al., 2024). In platforms managing high-value assets and physical cheque disbursements, embedding automated, unalterable database audit trails directly reduces internal security risks. Embedding this real-time data validation transforms software from a passive archive into an active risk-mitigation tool (Fatima Ezzahra et al., 2025). However, existing literature remains heavily conceptual or restricted to large-scale corporate manufacturing. The proposed system translates these compliance principles into a localized environment for Heindrich Trading Co. by hard-coding distinct user privilege boundaries and establishing an immutable system auditing layer to securely track all financial state changes and late-arriving customer payments.

Telemetric Networks, Distributed Ledgers, and Real-Time Asset Tracking

This is supported by Pethe et al. (2024), who evaluated telemetric asset-tracking networks using edge-to-cloud device gateways. Their framework successfully deployed automated notification scripts and threshold indicators to minimize logistical downtime. However, these existing frameworks depend on capital-intensive infrastructures, such as external RFID/GPS sensor hardware arrays or complex blockchain networks, making them inaccessible for small trading businesses. The proposed system addresses this limitation by deploying asynchronous background scripts specifically utilizing Celery task queues and Redis message brokers as a low-cost, pure-software alternative. Instead of using expensive external sensors, these background workers execute volume variance tracking natively through database constraints, instantly flagging transit discrepancies without slowing down the user interface.

Systematization and Meta-Analyses of Digital Transformation Trends

Macro-level academic evaluations establish clear baselines for digital system integration within the logistics sector. A meta-analysis of 1,264 academic papers mathematically validated that integrating IT-based supply chain management platforms directly enhances global transparency, inter-organizational communication, and sustainable business operations (Chiang et al., 2021). Structured coordination software consistently prevents operational friction, eliminates administrative backlogs, and reduces delivery delays across logistics channels (Reznik et al., 2022). Because these systematic reviews focus on global, macro-enterprise trends, they explicitly conclude by highlighting a critical need for localized empirical studies to test custom software configurations in real-world settings. Developing a custom management information system for Heindrich Trading Co. directly satisfies this requirement, serving as a localized empirical test case that applies these broad, systemic principles to optimize a small-scale commodity trading enterprise.

Table 2.1

References

Author & Date Published	**Description**	**Technology Features Used**	**Gaps in Research**
Wang (2021)	The study developed an intelligent data framework mapping information flow patterns across distributed enterprise networks. It explored the ways in which iterative waterfall network configurations resolve low informational awareness, loose business compliance, and fragmented logistical communication trails.	- Horizontal system integration - Localized sub-database clusters - Node optimization algorithms - Network time synchronization protocols - Distributed asset monitoring paths - Graphically displayed layout layers	This paper didn't consider what happens with these distributed nodes when they control certain retail activities, are involved in localized invoicing, or manually track goods in the warehouse within an independent distributor organization. The Heindrich Trading Co. software implementation addresses this apparent “gap” in conception by conducting a gap analysis of the manual procedure that is being repeated in the business.

Author & Date Published	Description	Technology Features Used	Gaps in Research
	Results have revealed that there is an improvement in data processing speeds and a marked optimization in distributed cluster synchronization when deploying algorithmic node mapping.		
Bhanu & Krithika (2025)	Proposed a structural framework verifying how information reliability scales organizational throughput via transparency metrics. Proved that content, scope, and data comprehensiveness	- Fuzzy Analytic Network Process (ANP) - Data Envelopment Analysis (DEA) - Dimensional transparency indexing (timeliness, regularity, scope)	Heavily dependent on macro-level analytics matrices. It bypasses low-level relational validation architectures required by independent commodity traders to track cross-referenced logistics expenses and real-time shipment discrepancies.

Author & Date Published	Description	Technology Features Used	Gaps in Research
	account for a 75% direct impact on supply chain transparency factors.		
Mirbagheri et al. (2023)	The study engineered a data-driven Decision Support System (DSS) utilizing a centralized data warehouse framework to maximize supply chain transparency. Mirbagheri's experimental findings demonstrated that unifying isolated legacy operations into structured storage clusters empowers managers to monitor	- Data-Driven Decision Support Systems (DSS) - Centralized data warehouse architectures - Extract, Transform, Load (ETL) tools - Real-time business intelligence dashboards - Historical trend analysis algorithms - Data profiling and cleansing protocols	A distinct limitation of this framework is its reliance on pre-existing enterprise data structures, bypassing the low-level relational database schemas and user-access configurations necessary for independent retail merchants.

Author & Date Published	Description	Technology Features Used	Gaps in Research
	real-time stock balances and detect seasonal distribution trends.		
Leong et al. (2024)	Evaluated how integrating strict business ethics and transparent supplier criteria into corporate operational workflows builds marketplace trust, reduces legal risk, and optimizes logistics sustainability.	- Immutable transaction recording layers - Multi-tier user privilege boundaries - Anti-corruption data architectures - Automated transparency dashboards - Audit trail mapping frameworks	Focuses heavily on high-level corporate policy guidelines, supplier screening strategies, and global environmental metrics. It provides no technical blueprints for lightweight web database constraints that address manual data-handling friction within small administrative teams.
Fatima Ezzahra et al. (2025)	Investigated the evolution of digital management systems, proving that transitioning away	- Real-time operational data pipelines - End-to-end operational visibility modules	Highly conceptual and enterprise-centric in scope. It does not detail the practical, browser-accessible database schemas or single-click user flows

Author & Date Published	Description	Technology Features Used	Gaps in Research
	from traditional, retrospective data filing into active validation layers converts software into a real-time risk mitigation tool.	- Dynamic performance metrics	needed by small trading firms to track fluid commodity shipments across third-party logistics networks.
Yu, X., & Zhou, Q. (2020)	Focuses on an ASP.NET-based web application designed to integrate macro-level supply chain layers (suppliers, manufacturers, distributors, and end customers) for real-time tracking and decision support.	- Browser/Server (B/S) and Client/Server (C/S) hybrid setups - Three-tier architecture (browser, application server, database server) - Distributed database support (MySQL, Oracle, SQL Server) - Electronic Data Interchange (EDI) and ERP integration	Focuses strictly on macro-enterprise architectures. It omits implementation data, financial cost barriers, and mobile-enabled deployment strategies for small, independent distributors.

Author & Date Published	Description	Technology Features Used	Gaps in Research
		- GPS/GIS tracking and barcode data collection	
Dong, X., & Salwana, E. (2022)	Analyzes the quantitative impact of migrating human resource and supply chain management data to centralized cloud repositories to optimize cross-departmental operations.	- Cloud-Based Supply Chain Management (CBSCM) - Cloud-Based Human Resource Management (CBHR) - Centralized database coordination paths - Cross-departmental digital workspaces	Assumes pre-existing corporate digital infrastructure. It fails to map specific database schemas or data entry transition models needed to convert manual paper records into digital formats.
Hijjawi, H., et al. (2023)	Investigates how digital supply networks act as a mediator to optimize operational performance and communication structures when	- Structural Equation Modelling (SEM) - SmartPLS analysis software - Digital supply chain integration frameworks	Evaluated exclusively within large-scale manufacturing firms. It fails to address localized inventory tracking, direct over-the-counter billing, or manual bookkeeping challenges common to small retail/wholesale businesses.

Author & Date Published	Description	Technology Features Used	Gaps in Research
	paired with lean management practices.	- Structural data communication dimensions	
Maheshwari, S., Gautam, P., & Jaggi, C. K. (2021)	Identifies security risks and information delays caused by traditional siloed data storage, proposing real-time analytical dashboards for proactive decision-making.	- Big Data Analytics (BDA) frameworks - Real-time analytical tracking tools - Automated data visualizations and dashboards - Predictive and descriptive diagnostic algorithms	Requires advanced, high-overhead big data engineering infrastructures. It completely skips basic web application configurations needed by standalone vendors to track stock without a dedicated data team.
Dashtizadeh, M., Meskaran, F., & Tan, D. (2022)	Details a secure, permissioned cryptographic ledger framework designed to record transaction histories and track	- Cryptographic hash functions and asymmetric-key encryption - Smart contracts via Solidity script execution	Centered heavily on specialized pharmaceutical supply chains (vaccine verification). It does not provide cost-benefit metrics or interface usability tests for low-budget, standard trading environments.

Author & Date Published	Description	Technology Features Used	Gaps in Research
	items across distribution networks.	- Decentralized ledger systems - React, HTML, CSS, and JavaScript front-end UI - Node.js server-side scripting	
Nofandi, F., Janaputra, M. R. B., & Pratama, R. A. (2025)	Engineered a web-based decentralized supply network configuration designed to combat institutional mistrust, transactional misinformation, and delayed operational reporting.	- Prototyping methodology - Distributed tracking ledgers - Real-time multi-actor sync (suppliers, processors, distributors)	Relies on decentralized blockchain ledger systems which impose prohibitive capital and configuration overheads for small-scale operations. It completely omits a simplified, software-driven data workspace for contract-to-expense matching

Author & Date Published	**Description**	**Technology Features Used**	**Gaps in Research**
Reznik, N., Rudenko, S., & Pylypchuk, K. (2022)	Maps out the conceptual foundations of integrated logistics systems, proving that coordinated digital networks prevent operational friction and reduce delivery delays.	- Integrated supply chain management frameworks - Process coordination modeling - Systematization and comparative analysis techniques	Entirely limited to theoretical and definitions-based analysis. It provides no actual system code, database frameworks, or empirical tests in a live business setting.
Smerichevska, S., Prodanova, L., & Yakushev, O. (2024)	Contrasts static document digitization with active automated circulation, demonstrating that automating operational files reduces human data-entry mistakes.	- Electronic document management networks - Internet of Things (IoT) warehouse tracking - Cloud infrastructure hosting (AWS)Wi-Fi and Bluetooth communication protocols	Focuses on capital-intensive automation equipment (robotics, AI, complex IoT). It omits technical blueprints for a standalone web app that merges daily point-of-sale over-the-counter billing with basic inventory counts.

Author & Date Published	**Description**	**Technology Features Used**	**Gaps in Research**
Pethe, S., et al. (2024)	Evaluates telemetric architectures that use automated threshold triggers and warning systems to minimize warehouse stockouts and administrative downtime.	- Real-time asset-tracking telemetric networks - Edge-to-cloud device gateways - Automated stock-level reorder notification scripts - Diagnostic analytics dashboards and relational data storage	Relies heavily on external physical hardware deployment (RFID tags, GPS nodes). This introduces cost and setup barriers for small businesses looking for a purely software-driven database solution.
Chiang, C.-T., Kou, T.-C., & Koo, T.-L. (2021)	Evaluates historical IT-based logistics trends across 1,264 academic papers to outline how digital platforms improve transaction transparency and	- Keyword analysis frameworks - Social network analysis tools - Sustainable Supply Chain Management (SSCM) tracking metrics	As a meta-analysis, it provides only generalized macro-level policy suggestions. It specifically calls for localized empirical studies to test custom web applications in small-scale business environments.

Author & Date Published	Description	Technology Features Used	Gaps in Research
	inter-organizational communication.		

Figure 2.1

Theoretical Framework

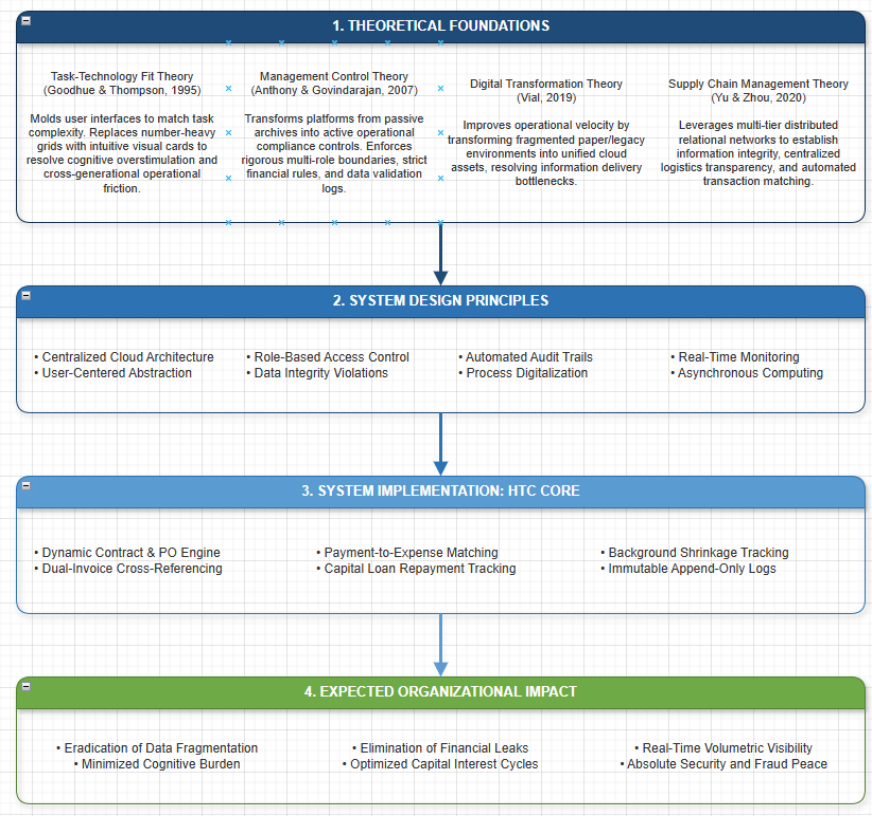


Figure 2.1 shows the theoretical framework of the study as a four-level flow.

Task-Technology Fit, Management Control, Digital Transformation, and Supply Chain Management theories inform the system design principles, such as centralized cloud architecture, role-based access control, automated audit trails, and real-time monitoring. These principles are applied in HTC Core through its contract and PO engine, dual-invoice cross-referencing, payment-to-expense matching, loan repayment tracking, shrinkage tracking, and immutable logs. The system is expected to reduce data fragmentation and financial leaks while improving visibility, security, and ease of use.

Synthesis

The reviewed literature establishes that migrating from paper-based, manual operations to cloud-hosted relational database environments is crucial for optimizing business performance, reducing administrative error, and mitigating operational risks (Yu & Zhou, 2020; Dong & Salwana, 2022; Fatima Ezzahra et al., 2025). Existing frameworks confirm that embedding algorithmic validation checks and automated threshold tracking into software workflows consistently eliminates transaction visibility gaps and delivery backlogs. (Fu et al., 2022; Reznik et al., 2022; Pethe et al., 2024). However, a distinct gap exists in current research: existing models focus primarily on macro-enterprise operations with large capital backing, or rely on expensive hardware infrastructure like RFID, GPS, and blockchain networks (Maheshwari et al., 2021; Dashtizadeh et al., 2022). These frameworks omit low-overhead relational schemas tailored for independent commodity distributors who must handle specialized needs like multi-modal barge logistics, transit-driven volume-to-density fluctuations, and complex payment-to-expense voucher matching (Yu & Zhou, 2020; Pethe et al., 2024). Furthermore, corporate platforms often overlook the usability constraints of non-technical administrative staff who require single-click user flows. Our custom system bridges these academic gaps by bypassing expensive physical tracking hardware. Instead, it embeds cross-departmental data abstraction, dynamic interest-accrual modeling, and unalterable system audit trails directly into pure software logic, serving as a localized empirical test case for independent, high-stakes distributors.

CHAPTER 3

METHODOLOGY

This chapter describes how the proposed cloud-based management information system for Heindrich Trading Co. will be planned, built, and validated. Based on stakeholder interviews, the company's core need is to replace manual Excel-based molasses trading records with a centralized web platform that supports contract and purchase order tracking, delivery monitoring (including delivered versus received quantities), barge and tracking service costs, sales invoice and cash voucher references, per-customer dashboards, and payment–expense matching for finance staff. The methodology aligns with the objectives and scope in Chapters 1 and 2: a cost-effective, browser-based system with multi-tier user access, without RFID/IoT hardware, public e-commerce, or full corporate accounting. HTC (Heindrich Trading Co. internal entity referenced in operations) is used in records for storage and related transactions; the proposed system serves the same organization.

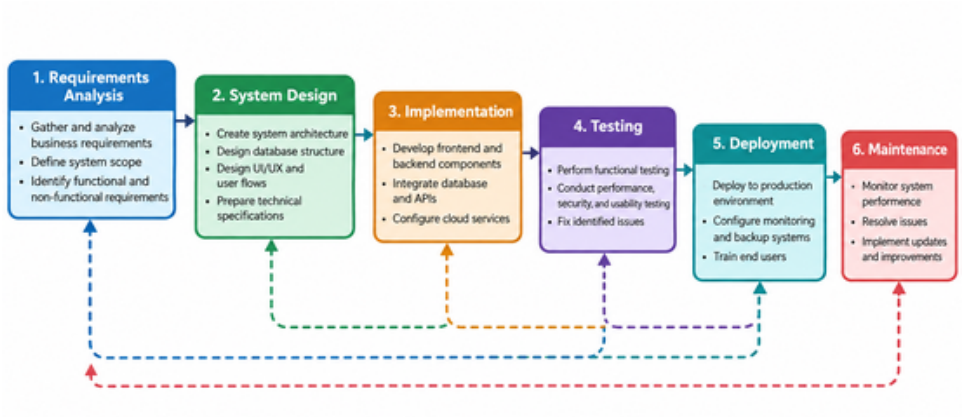
This chapter covers the development model, requirement analysis, hardware and software requirements, target users and environment, data gathering methods, and initial system design. Descriptive research was utilized to gather granular information regarding the company's existing workflows, transaction recording practices, inventory monitoring procedures, and financial tracking activities. While the core

framework focus remains developmental, descriptive research functions supplementarily to capture exact, real-world operational parameters that cannot be extracted from raw data models alone.

To achieve this, the researchers conducted structured and semi-structured qualitative interviews with key operations and finance employees within Heindrich Trading Co. Engaging directly with the staff members handling day-to-day data entry and fiscal reconciliations allowed the researchers to document exact pain points, identify where manual spreadsheet dependencies fail, and map empirical user requirements. The insights gathered from these personnel directly informed the relational database constraints and business logic rules necessary to establish an accurate operational framework.

Figure 3.1

Iterative Waterfall Model



The Iterative Waterfall Model serves as the framework for developing the proposed web-based supply chain management system for Heindrich Trading Co.

As illustrated in Figure 3.1, the model consists of six phases: requirements analysis, system design, implementation, testing, deployment, and maintenance. Each phase follows a structured sequence to guide the development process, while the feedback loops allow the researchers to revisit previous phases

when errors or necessary improvements are identified. This approach is suitable for the project because the proposed system integrates several business functions, including procurement, contract management, logistics cost computation, and reporting.

The first phase, Requirements Analysis, focuses on identifying the company's operational needs and documenting the system's required features. This is followed by System Design, where the database structure, user interface, and system architecture are planned. Mirbagheri et al. (2023) examined a data-driven Decision Support System that uses centralized data storage to improve supply chain transparency and operational monitoring. Their study highlights the importance of organizing business information to support data accessibility and informed decision-making. These considerations are relevant to the proposed system, which aims to consolidate procurement records, contract information, and logistics expenses within a unified platform.

The next phase, Implementation, involves developing the system based on the approved design, followed by Testing, which evaluates whether its features function according to the established requirements. Yu and Zhou (2020) discussed a web-based supply chain application that integrates information across different supply chain participants through web architecture and database technologies. Their study provides a relevant reference for developing a platform that connects related business information. In the proposed project, implementation and testing will focus on ensuring that user role management, procurement and contract records, logistics cost calculators, and reporting dashboards operate correctly and consistently.

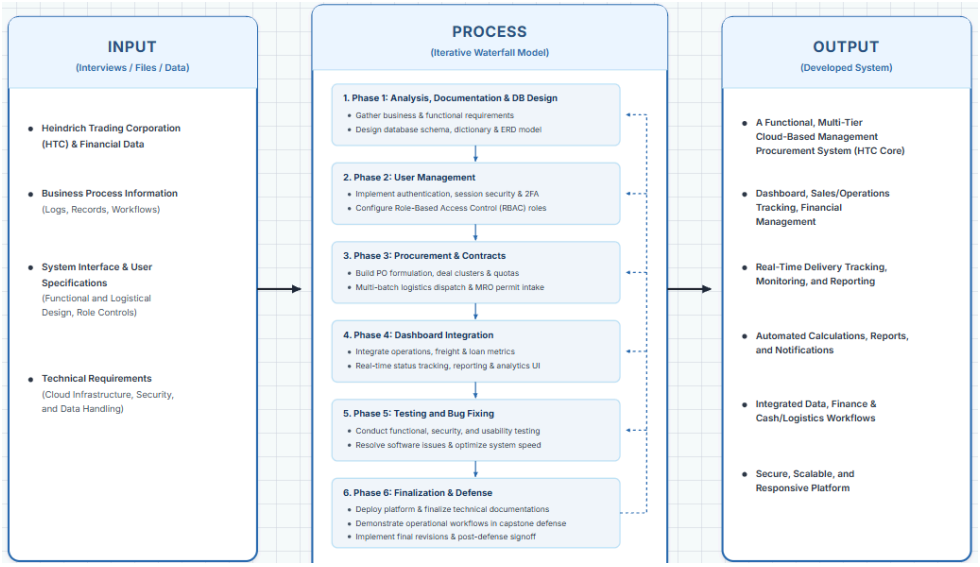
After testing, the system proceeds to Deployment, where it is prepared for use by authorized personnel of Heindrich Trading Co. The final phase, Maintenance, focuses on monitoring system performance, correcting errors, and implementing necessary improvements after deployment. Smerichevska et al. (2024) examined how automating document circulation can help reduce errors associated with manual

data entry. This is relevant to the proposed system because digitizing operational records may help minimize repetitive manual work and improve the consistency of business information management.

The feedback loops represented by the dashed arrows indicate that the development process is not strictly linear. When issues are identified during testing or actual use, the researchers may return to earlier phases to revise requirements, modify the system design, or correct implemented features. This allows the proposed system to be refined throughout development and after deployment. Overall, the Iterative Waterfall Model provides a systematic approach to developing a web-based solution that addresses Heindrich Trading Co.'s operational needs, while the reviewed studies highlight the importance of centralized information management, web-based system integration, and business process automation.

Figure 3.2

Input-Process-Output (IPO) Diagram



Input-Process-Output (IPO) diagram of the proposed cloud-based management and procurement system for Heinrich Trading Corporation (HTC).

The input consists of the business process information, financial records, stakeholder requirements, and technical specifications gathered to identify the organization's needs. These inputs serve as the basis for the system development process, which follows the Iterative Waterfall Model. This model consists of six phases: Requirements Analysis, System Design, Implementation, Testing, Deployment, and Maintenance. Each phase follows a structured sequence, while feedback loops allow necessary revisions and improvements when issues are identified. The output is the developed system, which aims to support the organization's management and procurement operations through integrated dashboards, sales and logistics tracking, automated calculations and reporting, financial management, and centralized data handling.

Table 3.1

Development Phases

Phase	**Deliverable**
Phase 1	Analysis, Documentation, DB Schema
Phase 2	User Management
Phase 3	Procurement & Contracts
Phase 4	Dashboard Integration
Phase 5	Testing and Bug Fixing
Phase 6	Finalization & Defense

The development of HTC Core is structured into six distinct phases designed to ensure systematic software development, rigorous technical validation, and seamless organizational adoption. As outlined in Table 3.1, these phases guide the transition of Heindrich Trading Corporation's operations from paper-reliant legacy workflows to a cloud-hosted enterprise platform:

Phase 1: Analysis, Documentation, and Database Schema Focuses on foundational architectural planning and requirements engineering. The team conducts stakeholder interviews and audits existing operational logs to draft functional specifications, define security boundaries, and model the PostgreSQL relational database schema.

Phase 2: User Management Establishes platform access governance, role-based access control (RBAC), and authentication frameworks. This phase builds secure login routes, user profile views, and permission boundaries across administrative, finance, invoicing, and operational user roles.

Phase 3: Procurement and Contracts Engineers the primary operational logic of the platform. It constructs core models for tracking upstream sugar mill contracts (e.g., BUSCO, CASA), downstream customer purchase orders, Molasses Release Order (MRO) permit management, and multi-modal barge/trucking logistics ledgers.

Phase 4: Dashboard Integration Consolidates isolated operational modules into unified visual dashboards. This phase integrates real-time status cards, payment-to-expense matching workflows, automated tax calculation tools, and Celery/Redis asynchronous workers for transit volume shrinkage calculations and dynamic daily interest tracking.

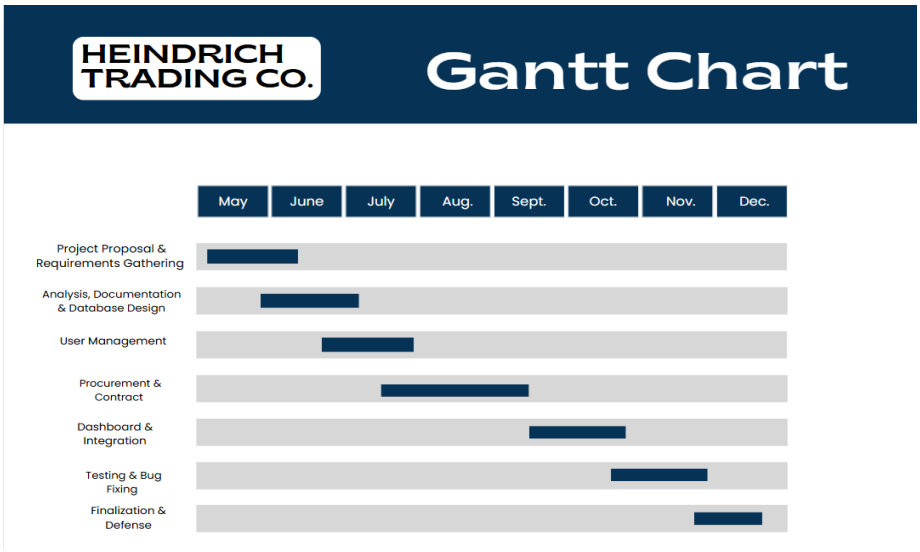
Phase 5: Testing and Bug Fixing Executes automated and end-user testing procedures. The system undergoes multi-role functional validation via Selenium WebDriver, load and latency

benchmarking, security vulnerability scanning, and User Acceptance Testing (UAT) with HTC management and staff.

Phase 6: Finalization and Defense Focuses on final production deployment on Amazon Web Services (AWS EC2), complete system documentation, user training sessions, and capstone academic defense presentation.

Figure 3.3

Gantt Chart



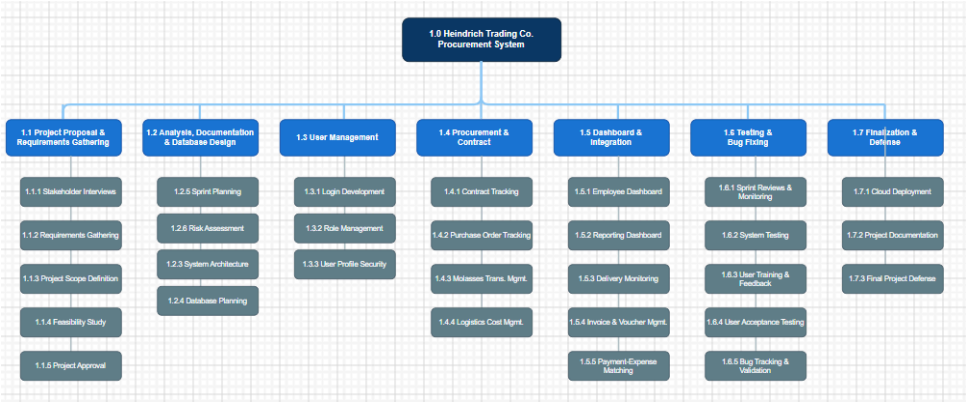
The proposed project follows an eight-month timeline spanning from May to December, transitioning systematically from descriptive research to iterative system development using the Waterfall framework.

The timeline begins with a two-month inception and architectural phase (May–June) dedicated to planning, stakeholder interviews, requirement documentation, and database modeling. This is followed by a four-month core development period (June–October) executing Phases 1 through 4, which systematically constructs user role management, the core molasses procurement and contract engine, logistics and barge

cost calculators, and integrated reporting dashboards. The timeline concludes with a two-month quality assurance and deployment phase (October–December) covering Phase 5 and 6 which includes system testing, bug fixing, official User Acceptance Testing (UAT) sign-off with Heindrich Trading Co. personnel, and final oral defense preparation.

Figure 3.4

Work Breakdown Structure



The Work Breakdown Structure (WBS) provides a deliverable-oriented hierarchical decomposition of the entire project scope into manageable, actionable work packages. Anchored by the main project node, 1.0 Heindrich Trading Co. Procurement System, the WBS organizes tasks across seven Level-1 branches:

1.1 Project Proposal & Requirements Gathering

Defines project feasibility and scope boundaries through direct field data collection:

1.1.1 Stakeholder Interviews: Conducting semi-structured interviews with HTC operations and finance personnel to map daily paper/Excel workflows.

1.1.2 Requirement Gathering: Collecting physical documentation, Cash Vouchers, Purchase Orders, and 2026 consolidated spreadsheets.

1.1.3 Impact Scope Definition: Establishing core functional boundaries, financial limits, and architectural constraints.

1.1.4 Feasibility Study: Evaluating technical, operational, and budget feasibility within the ₱10,000 allocation.

1.1.5 Project Approval: Securing formal sign-off from academic advisers and company stakeholders.

1.2 Analysis, Documentation & Database Design

Establishes the system's structural blueprint and data pipelines:

1.2.1 Sprint Planning: Structuring development sprints aligned with the 8-month timeline.

1.2.2 Risk Assessment: Identifying technical constraints, connectivity risks, and hosting availability considerations.

1.2.3 System Architecture: Designing the 3-tier presentation, processing, async, and storage layers.

1.2.4 Database Planning: Normalizing database entities, foreign keys, UUID clusters, and polymorphic audit logs.

1.3 User Management

Constructs security, authentication, and role governance:

1.3.1 Logic Development: Coding authentication backends, password hashing (PBKDF2), and session handlers in Django.

1.3.2 Role Management: Implementing granular Role-Based Access Control (RBAC) across 7 specialized role categories.

1.3.3 User Profile Views: Designing customized user profile navigation and setting controls.

1.4 Procurement & Contract

Builds core business operations and logistics tracking components:

1.4.1 Contract Tracking: Logging bulk molasses contracts, committed volumes, pricing, and fulfillment balances.

1.4.2 Purchase Order Tracking: Mapping customer purchase orders and multi-destination batch dispatches.

1.4.3 Release Orders (MRO) Management: Developing MRO permit linking and automated AWS Textract OCR parsing scripts.

1.4.4 Logistics Challenges / Expenses: Building barge and land transport cost calculators, 50% deposit trackers, and freight ledgers.

1.5 Dashboard & Integration

Merges backend processes into intuitive user interfaces and real-time computation engines:

1.5.1 Invoicing Dashboard: Constructing outward billing tools with automatic 12% Output VAT and 1%/2% EWT tax splits.

1.5.2 Reporting Dashboard: Generating operational summary reports, cash flow graphs, and contract status cards.

1.5.3 Delivery Monitoring: Displaying real-time shipment progress, loading/unloading weights, and transit status updates.

1.5.4 Contracts & Invoices Sync: Cross-referencing supplier invoice numbers against internal Cash Vouchers and Sales Invoices.

1.5.5 Payment System Matching: Implementing 3-step payment-to-expense allocation matching and active loan repayment engines.

1.6 Testing & Bug Fixing Ensures system stability, security compliance, and usability standards:

1.6.1 Safety Review & Monitoring: Assessing server-side security headers, CSRF tokenization, and input sanitization.

1.6.2 System Testing: Executing automated functional test suites using Selenium WebDriver and headless Python scripts.

1.6.3 User Training & Feedback: Orienting HTC personnel on dashboard interaction and ergonomics controls.

1.6.4 User Acceptance Testing (UAT): Administering ISO/IEC 25010 survey evaluations to end-users.

1.6.5 Bug Tracking & Validation: Resolving identified latency bottlenecks, UI errors, and software defects.

1.7 Finalization & Defense

Executes operational rollout and satisfies academic graduation deliverables:

1.7.1 Cloud Deployment: Deploying containerized Docker images to production AWS EC2 cloud infrastructure behind an NGINX reverse proxy.

1.7.2 Report Documentation: Compiling Chapters 1 through 5, system design diagrams, appendices, and source code listings.

1.7.3 Actual Project Defense: Presenting empirical evaluation findings and demonstrating system capabilities to the capstone examination panel

Table 3.2

Team Roles and Responsibilities

Member	Role	Responsibilities
Bernardino, Denielle	UI/UX Designer	Owns the end-to-end user interface and experience design; creates wireframes, prototypes, and visual assets.
Cahindi, Joseph Francis	Full-Stack Engineer	Collaborates with the team leader on technical architecture; drives both frontend and backend code implementation.
Maranda, Sam	Research Lead	Manages project documentation, academic writing, and research papers.
Santos, Kurt Zildjian	Project Leader & Backend Engineer	Oversees project milestones, finalizes deliverables, and manages database architecture and software.

Constraints

Technical Constraints

Internet Connectivity

The platform operates as an AWS EC2 cloud-hosted web application requiring continuous network connectivity for real-time relational data synchronization, contract logging, and financial state updates. The platform is engineered around a target SLA latency threshold of 1,000 ms and a maximum page load time under 1.2 seconds. Performance benchmarks indicate that under a concurrent load of 10 virtual users generating 200 total requests across core application routes, system throughput achieves 5.02 requests/second with an average API response latency of 1,366.75 ms. Network degradation dropping available bandwidth below 1.5 Mbps or elevating latency beyond 2,000 ms results in temporary UI state synchronization delays

Browser Compatibility

HTC Core is optimized for modern web browsers compliant with HTML5, CSS3, and ECMAScript 2020+ standards, specifically Google Chrome (v100+), Microsoft Edge (v100+), Mozilla Firefox (v100+), and Apple Safari (v15+). The presentation tier utilizes Bootstrap 5, requiring client engines capable of rendering flexbox and grid layouts at minimum viewport widths of 320px (mobile/tablet) to 1920px (desktop). To support accessibility and senior ergonomics, the interface enforces strict touch-target dimensions of 48px and high-contrast font scaling, which may exhibit visual distortion or reduced interactivity on legacy browsers lacking modern CSS variable support.

Cloud Server Availability

Since the platform relies on cloud infrastructure for data storage and application hosting, system availability depends on the uptime and reliability of the hosting provider. Scheduled maintenance, server interruptions, or unexpected outages may temporarily affect access to the system.

Data Storage Capacity

Transactional data, contracts, PO confirmations, invoices, logistics logs, and immutable audit trails are stored within a centralized PostgreSQL relational database engineered to scale up to 500,000+ delivery records and support 50+ concurrent internal operators. High-precision fixed-point numeric formats are enforced across all financial and volumetric schema fields to eliminate rounding errors over cumulative datasets. Unstructured document uploads (PDF scan outputs processed via AWS Textract OCR) are isolated in dedicated cloud media storage to prevent primary database bloating and maintain sub-second relational query indexing as historical logs expand.

Non-Technical Constraints

User Adoption and Training

The operational personnel of Heindrich Trading Co. currently rely on spreadsheet-based workflows and manual documentation processes. Transitioning to a centralized management information system may require user orientation and training to ensure proper utilization of system features and workflows.

Development Timeline

The project follows the Iterative Waterfall Model and is constrained by the Capstone 1 and Capstone 2 academic schedule spanning an eight-month delivery timeframe from May to December.

Development activities, including requirement gathering, system design, implementation, testing, and deployment, must be completed within this allotted academic timeframe.

Data Confidentiality

Heindrich Trading Co. handles sensitive procurement contracts, supplier pricing information, customer transactions, and financing records. Due to the confidential nature of these business operations, access to certain data must remain restricted to authorized personnel and may limit the amount of real-world financial information available during system development and testing.

Organizational Scope

The proposed system is designed specifically for the internal operational requirements of Heindrich Trading Co. The platform focuses on procurement management, logistics monitoring, invoicing, and payment-to-expense matching. Functions such as full accounting, tax management, regulatory compliance automation, public customer portals, and e-commerce services are outside the scope of the project.

Resource and Budget Limitations

As an academic capstone project, the development team operates under a capped initial working budget of ₱10,000.00 (Ten Thousand Philippine Pesos) to cover all operational, technical, and administrative expenses. This fixed financial allocation is specifically distributed across four primary expenditure categories:

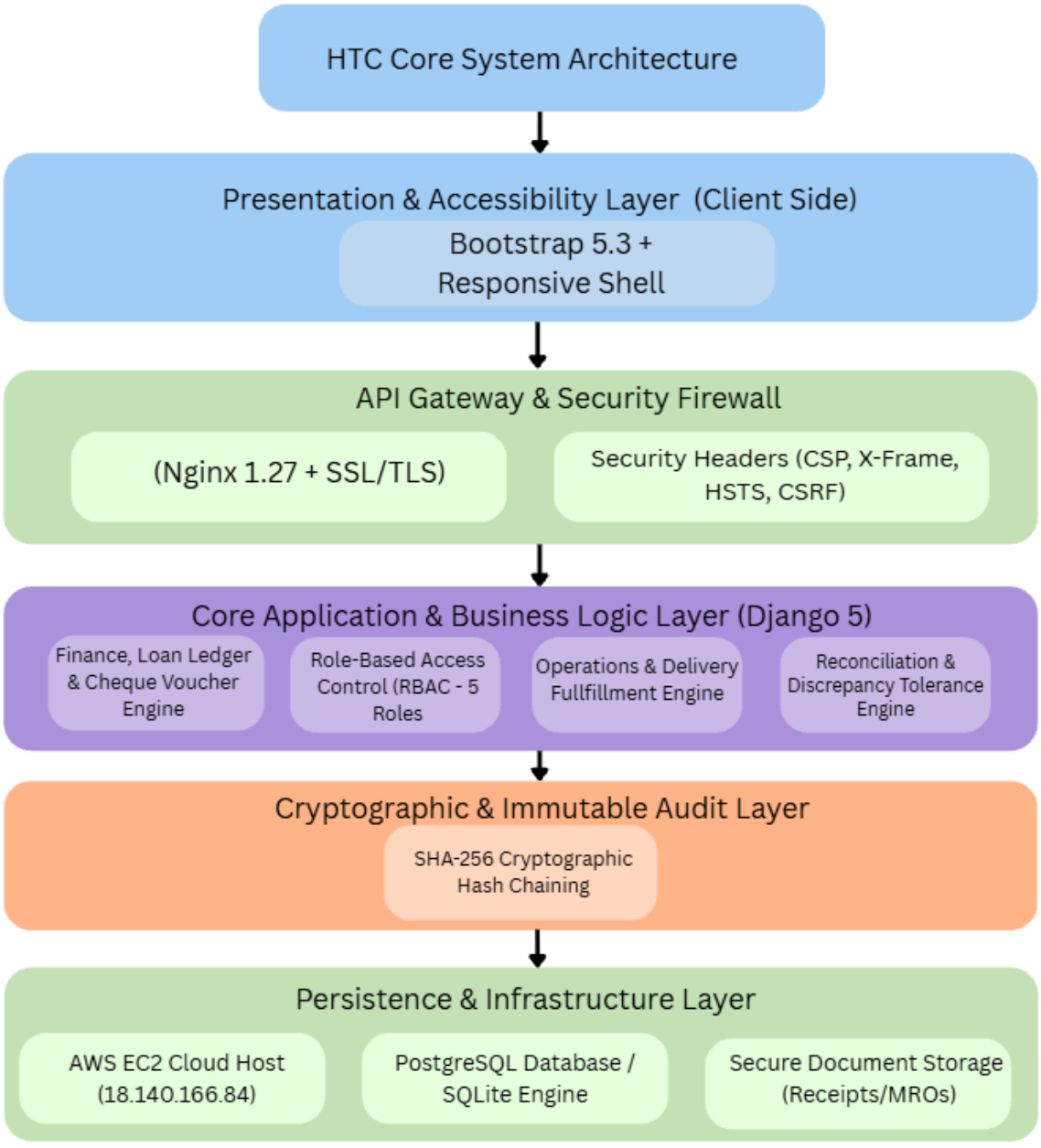
1. Domain Name Registration: Securing a dedicated domain for web platform deployment and user access.
2. Cloud Server & Infrastructure Hosting: Funding cloud compute instances, database storage, and container runtime on Amazon Web Services (AWS EC2).

3. Agentic Tokens & API Consumption: Purchasing API token usage for automated document parsing via AWS Textract OCR and interactive AI assistant queries (HTC Copilot).
4. Project Documentation & Academic Requirements: Printing, manuscript binding, survey instruments, and administrative paperwork required for capstone evaluation and defense.

To maintain strict adherence to this ₱10,000.00 limit, the system architecture relies exclusively on open-source technologies—including Django, PostgreSQL, Celery, Redis, Bootstrap 5, and Docker—while deliberately excluding capital-intensive hardware arrays such as RFID tags, barge GPS units, and proprietary enterprise ERP licenses

Figure 3.5

System Architecture



1. Presentation & Accessibility Layer (Client Side)

The Presentation & Accessibility Layer serves as the frontline client interface engineered to provide an intuitive, responsive, and accessible experience for operational and executive personnel across diverse devices. Implemented with Bootstrap 5.3 and custom styling within templates/base.html and module-specific templates, this layer strictly adheres to Web Content Accessibility Guidelines (WCAG 2.1 Level AA). It incorporates minimum color contrast ratios of 4.5:1 for standard text (elevating beyond 7.0:1 in High-Legibility Mode), accessible :focus-visible keyboard focus indicators, a root <a href="#mainContent" class="skip-to-content"> bypass link for keyboard and screen-reader navigation, and interactive touch targets exceeding the 44×44x44×44 pixel standard. A fluid, responsive grid paired with a collapsible sidebar (sidebar-collapsed) guarantees seamless usability across desktop terminals, tablets, and mobile smartphones in sugar mill weighbridges and port terminals. Furthermore, this layer hosts client-side input guardrails, statistical irregular mean alerts (such as flagging single truck volumes exceeding 60.0 MT), and dual-step confirmation modal dialogs that intercept accidental user errors before data is submitted over the network.

2. API Gateway & Security Firewall Layer

The API Gateway & Security Firewall Layer acts as the secured network perimeter defending the internal application server against external web threats while optimizing transport performance. Powered by an NGINX 1.27 reverse proxy hosted on AWS EC2, this layer terminates SSL/TLS 1.3 cryptographic sessions and directly serves compiled static assets (CSS, JS, web fonts) and validated media files, eliminating unnecessary load on backend Python workers. It enforces enterprise security headers across every HTTP response, including Content Security Policy (CSP), clickjacking prevention via X-Frame-Options: DENY, X-Content-Type-Options: nosniff, and HTTP Strict Transport Security (HSTS). NGINX also applies HTTP request rate-limiting, buffers network traffic, and works in tandem with Django's CsrfViewMiddleware to mandate cryptographically unique CSRF tokens on all data-mutating requests

(POST, PUT, DELETE). Validated HTTP traffic is forwarded through an isolated Unix domain socket to the Gunicorn WSGI Application Server, which manages multi-threaded worker pools executing the core application code.

3. Core Application & Business Logic Layer (Django 5)

The Core Application & Business Logic Layer represents the computational heart of HTC Core, engineered in Django 5 and Python 3.12 and structured into distinct modular applications: accounts, finance, operations, and audit. This layer houses four critical sub-engines:

Role-Based Access Control (RBAC): Centrally defined in accounts/permissions.py, RBAC enforces authorization across five discrete personas (Administrator, Operations Management, Operations, Finance, Invoicing) using @role_required view decorators and server-side template conditionals that suppress unauthorized controls.

Finance, Loan Ledger & Cheque Voucher Engine: Centered in finance/views.py, this engine automates commercial bank credit facilities, executes dynamic daily interest accrual algorithms, manages the 50% upfront hauler downpayment policy, and enforces the append-only "Void & Reissue" compensating cheque workflow (compensating_correction_voucher).

Operations & Delivery Fulfillment Engine: Managed in operations/views.py, this engine monitors bulk molasses contract sourcing, purchase order clusters, and multi-batch dispatch schedules with strict date-ordering validations.

Reconciliation & Discrepancy Tolerance Engine: This engine programmatically calculates shipping shrinkage (Volume Variance (MT) = Dispatched Metric Tons − Received Metric Tons

Shrinkage Percentage (%) = [(Dispatched Metric Tons − Received Metric Tons) / Dispatched Metric Tons] × 100) 1.00% loss threshold are locked in a DISPUTED_VARIANCE state, blocking final customer billing until managerial sign-off. Intensive background recalculations and nightly interest updates are

offloaded to Celery 5 distributed workers backed by a Redis 7 in-memory message broker, preserving sub-1.2-second synchronous page latency.

4. Cryptographic & Immutable Audit Layer

Situated as an unyielding security barrier between application logic and physical database storage, the Cryptographic & Immutable Audit Layer guarantees absolute transactional integrity, non-repudiation, and fraud prevention through SHA-256 sequential cryptographic hash chaining. Implemented in operations/services/hash_chain.py and finance/services/finance_hash_chain.py, every major operational milestone—from PO Capital Formulation (Genesis Block 0000...0000) through Bank Loan Authorization, Cheque Voucher Issuance, Weighbridge Receipt, to Final BIR Tax Invoicing—constructs a normalized, deterministic JSON payload. The block computes a SHA-256 hash that encapsulates its current data combined with the cryptographic digest of its immediate predecessor:

$$H_i = SHA - 256(D_i \parallel T_I \parallel H_{\{i-1\}})$$

At the data model tier, audit/models.py defines the SystemAuditTrail, which explicitly overrides Django's internal ORM save() and delete() methods to throw runtime ValueError exceptions if an in-place UPDATE or DELETE is executed. Any direct database tampering or manual SQL manipulation alters individual record values and breaks the sequential cryptographic digest chain, which automated verification algorithms detect immediately (is_intact: False).

5. Persistence & Infrastructure Layer

The Persistence & Infrastructure Layer forms the underlying physical and cloud foundation that guarantees high availability, strict ACID transaction compliance, and durable file storage. Hosted on an Amazon Web Services (AWS) EC2 Ubuntu Cloud Host (Elastic IP 18.140.166.84), it operates containerized runtime services managed under Docker and systemd. Primary relational data is persisted within an

enterprise PostgreSQL 16 relational database engine (with SQLite3 support for isolated testing). Database-level data integrity is fortified by the PostgreSQL prevent_audit_tampering() BEFORE UPDATE OR DELETE database trigger and foreign key ON DELETE RESTRICT rules that block unauthorized record destruction independently of application code. To prevent IEEE-754 floating-point rounding drift, the database schema enforces arbitrary-precision fixed-point fields: DECIMAL(12,2) for financial values and DECIMAL(14,3) for volumetric tonnages. Finally, a Secure Document Storage Subsystem manages digitized paper assets (Sugar Regulatory Administration MRO clearances, laboratory Brix test sheets, and official weighbridge tickets) stored within structured media volumes complete with MIME-type validation, file size bounds, and virus checks.

Table 3.3

Frontend Framework

Frontend Framework	**Language**	**Performance**	**Ease of Use**	**Project Suitability**
Bootstrap 5	HTML, CSS, JavaScript	Lightweight	High (Pre-built)	Optimal: Drastically cuts development time
React.js	JavaScript	Virtual DOM (High state-change speed)	Low (Requires custom styling/logic)	Overkill: Unnecessarily complex for current scope

Vue.js	JavaScript	Virtual DOM (High state-change speed)	Low (Requires custom styling/logic)	Sub-optimal: Requires extra architecture setup
Angular	TypeScript	Real DOM (High memory load)	High (Monolithic framework assets)	Unsuitable: Excessive Soverhead for project scope.
Tailwind CSS	HTML, CSS, JavaScript	Atomic CSS (Smallest file footprint)	Low (0% Unstyled utility classes)	Sub-optimal: Increases UI building time

This criterion assesses component readiness and architectural performance to determine suitability for the Management Information System (MIS). Bootstrap 5 offers a high component readiness level by providing 100% pre-designed UI elements (tables, buttons), whereas Tailwind CSS requires manual, utility-class configuration for every design element, significantly inflating development hours. Architecturally, while React and Vue leverage a Virtual DOM for dynamic UI state-changes and Tailwind utilizes Atomic CSS to minimize style sheets, these paradigms do not align with the system's primary requirement. Because the MIS prioritizes rapid initial page load times over hyper-reactive interface updates, Bootstrap's lightweight CSS/JS bundle is optimized to meet the platform's specific performance needs.

Table 3.4

Backend Framework

Backend Framework	**Language**	**Security & ORM**	**Native Admin & Auth Tools**	**Project Suitability**

Django	Python	Native SQLi/XSS protection	Built-in Admin & Auth	**Highly Suitable:** Built-in RBAC, Auditing & Logs
Laravel	PHP	Eloquent ORM	Partial (Requires separate packages)	**Lacks Native Tools:** Requires more setup than Django
Express.js	JavaScript	None (Requires 3rd-party modules)	None (Must code Auth from scratch)	**High Effort:** Requires third-party libraries
Spring Boot	Java	Hibernate ORM	None (Heavy custom configuration)	**Excessive:** Too heavy for system scope
Flask	Python	None (Requires SQLAlchemy extension)	None (Micro-framework structure)	**Inadequate:** Lacks built-in MIS features

This assessment evaluates the architectural security features of development frameworks upon initial installation, specifically contrasting Django's "batteries-included" paradigm with minimalist frameworks like Express.js and Flask. Django natively delivers integrated administration dashboards, user authentication

systems, and a robust Object-Relational Mapper (ORM) that automatically sanitizes inputs to mitigate SQL injection risks out-of-the-box. Conversely, Express.js and Flask lack native security infrastructure and database abstraction layers, necessitating manual, line-by-line configuration that introduces potential vulnerability risks. Consequently, Django's native, pre-configured security components provide a more resilient environment for safeguarding the transactional data logs of Heindrich Trading Co.

Table 3.5

Cloud Service Evaluation

Cloud Service	**Description**	**Advantages**	**Limitations**	**Suitability for HTC System**
Amazon EC2 (AWS)	Virtual cloud server that allows deployment of custom web applications using Django, PostgreSQL, Gunicorn, and NGINX.	Full server control, scalable resources, strong security features, supports custom software stacks, pay-as-you-use pricing.	Requires basic server administration knowledge.	**Highly Suitable.** Supports the proposed architecture (Django, Gunicorn, NGINX, PostgreSQL), provides secure cloud hosting, and allows future

				system expansion.
Microsoft Azure Virtual Machines	Cloud-based virtual servers for hosting enterprise applications.	Strong integration with Microsoft products and enterprise environments.	Higher complexity and potentially higher costs for small projects.	Suitable but may be excessive for a small-scale commodity trading company.
Google Cloud Compute Engine	Infrastructure-as-a-Service (IaaS) platform for deploying custom applications.	High performance and flexible configurations.	Requires additional configuration and management.	Suitable but provides fewer educational resources and community support compared to AWS.
Heroku	Platform-as-a-Service (PaaS) for deploying web applications.	Easy deployment and minimal server management.	Limited control over infrastructure and higher costs as usage increases.	Suitable for prototypes but not ideal for long-term operational deployment.

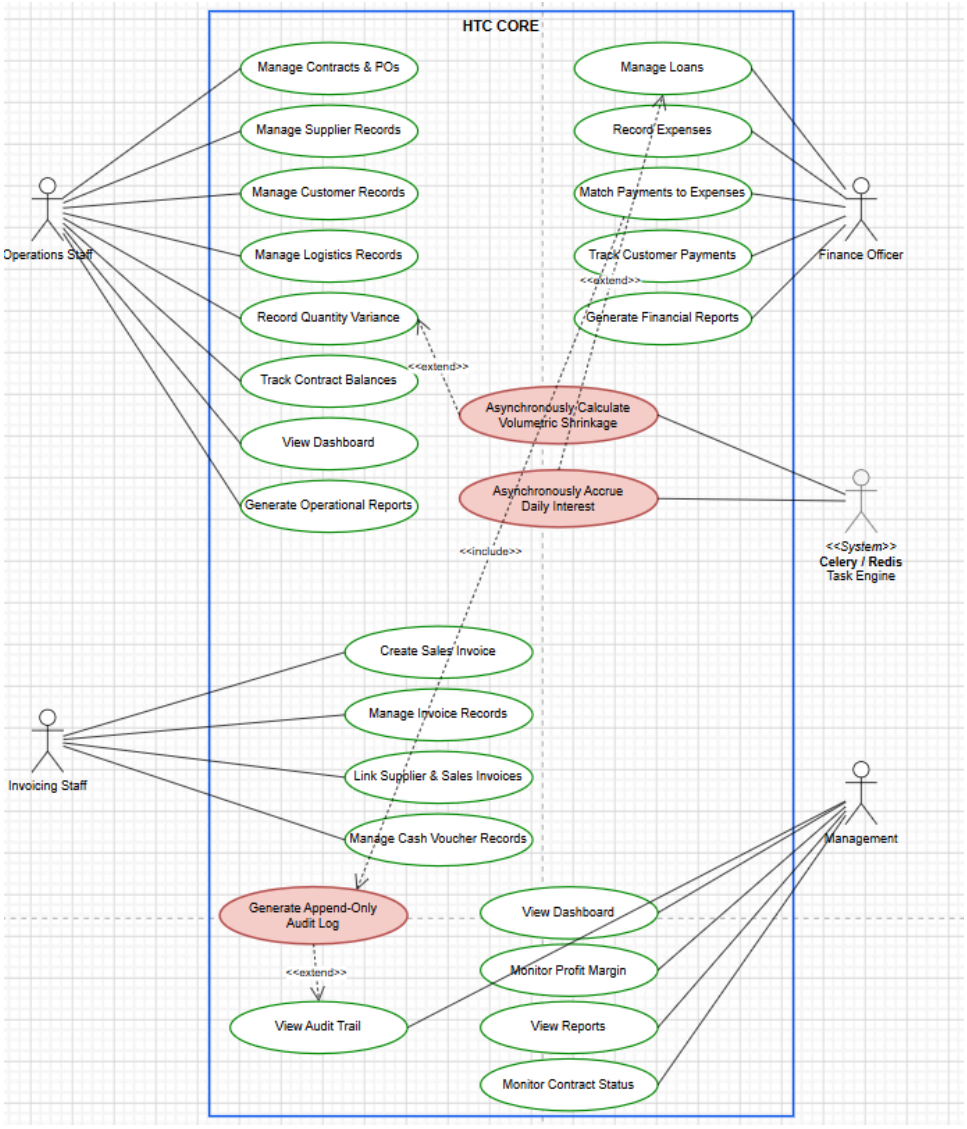
DigitalOcean Droplets	Simple cloud virtual machines for hosting web applications.	Affordable pricing and easy setup.	Fewer enterprise-level services and scalability options.	Suitable for small projects but less scalable than AWS EC2.

Table 3.5 presents an evaluation of different cloud hosting services based on their descriptions, advantages, limitations, and suitability for the proposed HTC System.

The comparison considers factors such as scalability, cost, infrastructure control, security, and ease of deployment. Among the evaluated services, Amazon EC2 (AWS) is identified as the most suitable option because it supports the system's required technologies, provides flexibility and security, and allows future expansion.

Figure 3.6

Use Case Diagram

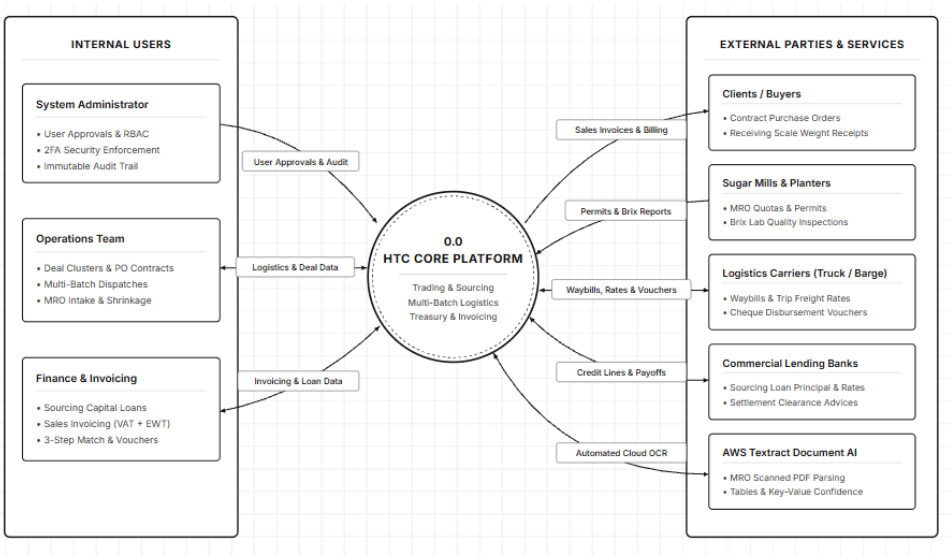


This architectural integration directly bridges human operational workflows with an automated backend task engine to enhance both application performance and ledger data integrity. When the Operations Staff records variance metrics or a Finance Officer manages a loan, the system utilizes an «extend» relationship to hand off heavy computations to a background worker powered by Celery and Redis. This prevents synchronous blocking of the user interface, allowing volumetric shrinkage tracking and

daily interest accruals to execute asynchronously without causing system latency for the user. Simultaneously, critical financial actions—such as matching payments to expenses—utilize an «include» relationship to automatically trigger an immutable, append-only system audit log task. This automated data pipeline ensures that the physical database layers stay synchronized and secure, directly feeding data to the real-time dashboards where management can monitor gross operational margin tracking, review immutable audit trails, and track contractual statuses with complete transparency.

Figure 3.7

Data Flow Diagram

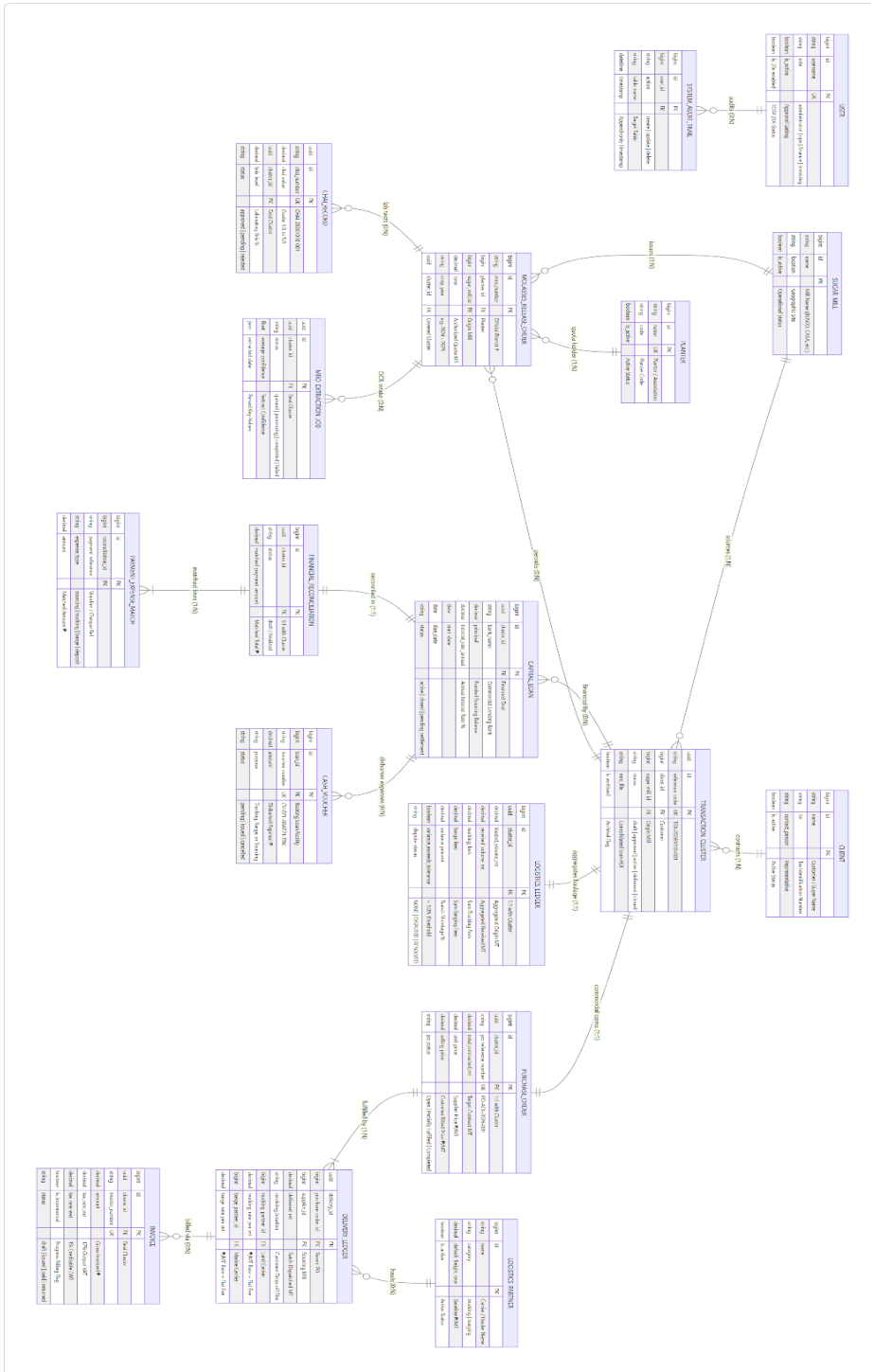


Data flows sequentially through five core processes to transform raw procurement inputs into actionable system alerts and role-specific metrics. Process 1.0 logs baseline supplier agreements into the Contract Database (D1), which Process 2.0 then pulls to handle asynchronous weight verification and shrinkage matrix generation within the Logistics Database (D2) without delaying the user interface. In parallel, Process 3.0 commits billing records to the Invoice Database (D3). Process 4.0 ingests these invoices and active loan parameters to manage total liabilities within the Finance Database (D4)—which

separates tentative drafts from finalized states—while silently exporting cryptographic log streams to the Immutable Audit Logs Database (D5). Finally, Process 5.0 aggregates data from D4 and D5 to deliver tailored operational intelligence: transit risk alerts for Operations Staff, live interest countdowns for the Finance Department, and streamlined gross operational margin overviews for Management, clarifying that these are calculated purely as contract revenue minus procurement and logistical transport costs.

Figure 3.8

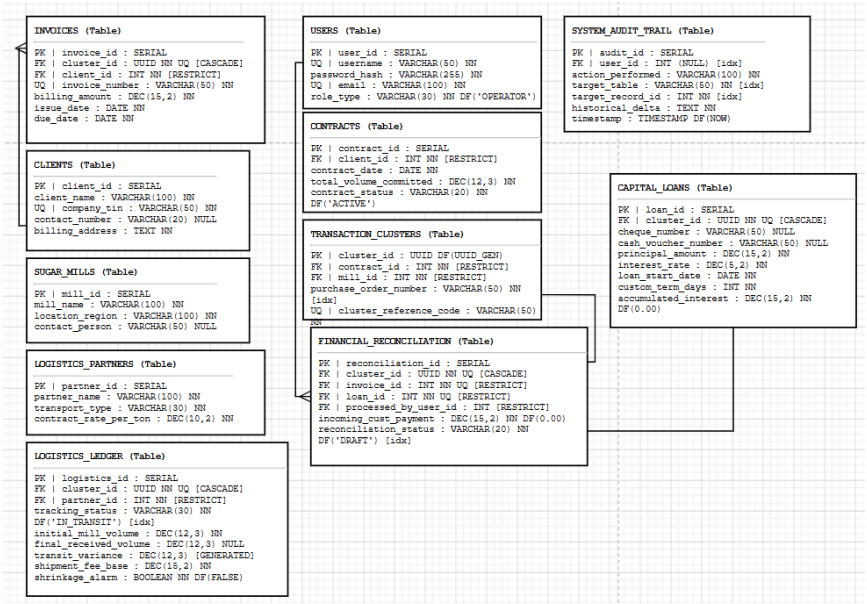
Entity Relationship Diagram (ERD)



This diagram separates master registries from operational data to maintain relational integrity and eliminate redundant data entry. Master tables (CLIENTS, SUGAR_MILLS, and LOGISTICS_PARTNERS) centralize core partner metrics, ensuring uniform TINs, addresses, and freight rates across the system. At the center of the operational core, TRANSACTION_CLUSTERS links customer purchase orders and supplier contracts to all downstream tracking modules. Physical inventory risks are managed via the LOGISTICS_LEDGER through automated transit variance calculation, while financial exposure is monitored in real time via the CAPITAL_LOANS dynamic interest engine. For compliance and security, INVOICES cross-reference transactions directly to bypass intermediate queries, FINANCIAL_RECONCILIATION enforces strict state-driven approval workflows, and a decoupled SYSTEM_AUDIT_TRAIL captures append-only historical deltas using a scalable polymorphic design.

Figure 3.9

Database Schema



The physical database schema is structured into distinct functional layers to ensure strict relational integrity, optimize query performance, and eliminate data redundancy. Foundational master registries consisting of CLIENTS, SUGAR_MILLS, and LOGISTICS_PARTNERS centralize core entity profiles, ensuring that uniform billing, geographical data, and contract freight rates remain consistent across all operations. At the heart of the transactional layer sits the TRANSACTION_CLUSTERS table. Utilizing unique UUID identifiers, this table serves as a centralized hub that binds upstream supplier contracts directly to individual sub-modules via rigid $1:1$ constraints, simultaneously mapping real-time cargo tracking details in the LOGISTICS_LEDGER and credit exposure in the CAPITAL_LOANS engine.Financial oversight, system security, and data accountability are managed through state-driven workflows and decoupled logging mechanisms. The FINANCIAL_RECONCILIATION table acts as the final gatekeeper for transaction processing, enforcing sequence control by tracking entries through a strict workflow that separates tentative draft statuses from finalized, audited accounting records. To capture a complete historical narrative of the database without degrading performance, the SYSTEM_AUDIT_TRAIL table implements a polymorphic architecture. By recording generic table-string names and corresponding record IDs rather than relying on standard database foreign keys, this append-only tracking engine captures absolute, unerasable historical deltas across every operational table while ensuring the underlying storage engine remains fully scalable as the application grows.

Data Gathering Procedures

The researchers gathered the information required for the development of HTC Core through direct stakeholder interaction, review of existing company records, and analysis of the organization's current operational workflow. The data gathering process focused on understanding how Heindrich Trading Corporation records procurement transactions, monitors deliveries, manages logistics expenses, tracks invoices and payments, and monitors financial obligations.

A descriptive research approach was used to document the existing business processes and identify the operational requirements of the proposed system. Structured and semi-structured interviews were used to obtain information from personnel involved in the company's daily operations and financial-related activities. The information gathered was used to identify existing problems, determine the required system functions, and establish the business rules to be implemented in HTC Core.

The researchers also reviewed existing company records, particularly the consolidated Excel records provided by the organization. These records were examined to understand the structure and types of information currently being recorded, including transaction details, purchase and selling prices, customer and supplier information, transportation services, invoice references, quantities delivered and received, and profit-related information. Appendix B contains the 2026 consolidated Excel sheet submitted by the company, subject to the company's right to redact confidential or proprietary information.

The gathered information served as the basis for the system requirements, database structure, user roles, transaction workflows, financial tracking functions, logistics monitoring, and dashboard requirements of HTC Core. The existing methodology of the study already identifies stakeholder interviews and operational records as sources for determining the company's actual workflow and system requirements.

Data Collection Flow

The data collection process followed a systematic sequence to ensure that the information used in the development of HTC Core was based on the organization's actual operational requirements.

First, the researchers identified the major business processes that needed to be studied, including procurement, contract and purchase order management, logistics monitoring, invoicing, payment tracking, and financial reconciliation.

Second, the researchers conducted interviews with relevant company personnel to understand the current procedures, problems, and information requirements. The interview focused on how transactions are recorded, monitored, communicated, and reconciled within the organization.

Third, the researchers reviewed available company records and the 2026 consolidated Excel file to identify the types of data currently maintained by HTC. The records were examined to determine the information fields and relationships necessary for the proposed database.

Fourth, the collected information was analyzed and translated into system requirements. These requirements were then used in the design of the database, user roles, business processes, dashboards, and system functions.

Finally, the identified requirements were incorporated into the iterative development process. The system components were developed through iterative phases, allowing the development team to refine the system based on identified requirements and stakeholder feedback. The existing development pllan begins with requirements gathering and analysis before proceeding to database design, user management, procurement and contracts, dashboard integration, testing, and finalization.

Ethical Considerations

Carrying out a research study and implementation of a software program in an operational business environment demands strict observance of ethics related to data protection and participant consent.

Interview / Existing Records

For a clearer view of the interviews we conducted with our stakeholders, and any other existing records relevant to the project, please refer to Appendix A onwards.

Sampling Method

The study employed purposive sampling to select key personnel with direct operational knowledge of the company. For qualitative data gathering, Ms. Jewel Marie A. Perez was selected as an information-rich interviewee to detail the existing transaction workflows, Excel-based record keeping, logistics coordination, and operational bottlenecks. For system evaluation, four (4) of the company's six (6) total personnel were selected, representing all administrative, operational, invoicing, and finance roles who directly interact with the deployed web platform. The remaining two (2) employees were deliberately excluded as their duties are limited to field messaging and do not involve web system operation.

Table 3.6

Participants/Respondents

Name	Job Description
Marinic A. Perez	Operations Manager
Jewel Marie A. Perez	Operations
Diane Peralta	Invoice
Digna Guizano	Finance

Inclusion Criteria

Participants were considered eligible for the study if they met the following criteria:

1. They were personnel of Heindrich Trading Corporation or were directly authorized to provide information regarding its business processes.

2. They had direct knowledge or involvement in operational activities relevant to the proposed system.
3. They were familiar with one or more of the following processes: procurement, contracts, logistics, invoicing, payment monitoring, financial reconciliation, or transaction recording.
4. They were willing to provide information necessary for requirements gathering and system development.
5. The information they provided could be used for academic system analysis and development while respecting company confidentiality.

Exclusion Criteria

The following were excluded from the data-gathering activity:

1. Individuals without direct knowledge of HTC's operational processes.
2. Information unrelated to the objectives and scope of HTC Core.
3. Confidential or proprietary company information that HTC did not authorize the researchers to access or use.
4. Personal information that was not necessary for identifying system requirements.
5. Business records or processes outside the scope of the proposed management information system.

Instruments and Tools Used

The research involved the use of research instruments, software platforms, and cloud-based technology to obtain operational data and assess the quality of the platform.

Research Instruments

1. Semi-Structured Interview Guide: A set of targeted questions used during a one-on-one interview with the Operations Staff (Ms. Jewel Marie A. Perez). It captured details on daily Viber communications, Excel tracking, 50% barge downpayments, bank loan tracking, and transit shrinkage losses (refer to Appendix A).

2. System Requirements Checklist: An auditing checklist used to review existing paper records (Cash Vouchers, Purchase Orders, Sales Invoices, Supplier Invoices) and Excel files. This ensured all necessary parameters—such as Cash Voucher numbers, 1% volume tolerance limits, and loan interest variables—were mapped into the database schema.

3. ISO/IEC 25010 Evaluation Questionnaire: A 5-point Likert scale questionnaire given to end-users and IT evaluators to assess seven software quality areas: *Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security,* and *Maintainability*.

4. Functional Test Cases: Automated testing scripts containing specific inputs, expected results, and pass/fail conditions to verify user logins, dynamic interest accrual calculations, asynchronous volume calculations, and audit logging.

Testing Procedures

The operational readiness, performance, and security of the HTC Core Enterprise Trading Platform, deployed on Amazon EC2, were evaluated through three pillars of automated testing: Functional Testing, Performance Testing, and Security Testing. Automated testing was employed to ensure the reliable execution of critical business workflows, the accurate enforcement of financial calculations and validation

rules, and the stability of the platform under concurrent user loads. Selenium WebDriver was used for end-to-end functional testing, a multithreaded Python engine was utilized for performance and load testing, and an automated security assessment suite was developed to evaluate common vulnerabilities based on OWASP security principles. Together, these testing methods provide a systematic and repeatable approach to assessing the platform's functionality, performance, and security.

Table 3.7

Functional Testing Matrix

Testing Type & Tools	**Test Scope**	**Results & Outcomes**
Functional Testing (Selenium WebDriver, Python Engine)	10-Stage Core Company Process (Multi-role auth, PO creation, Dual-approval, Math rule enforcement, Logistics variance, Sales invoicing, Payment reconciliation, Loan closure, Boundary validation)	**Pass Rate:** 100% (10/10 criteria). Business logic adhered to. Loan validation correctly blocked underfunded transactions.

Table 3.8

Performance & Load Testing Matrix

Testing Type & Tools	Test Scope	Results & Outcomes
Performance Testing (Concurrency Engine, ThreadPoolExecutor, Gunicorn telemetry)	10 Concurrent Virtual Users (200 requests total) across critical routes (Dashboard, Analytics, Operations, Finance)	**Success Rate:** 100%. Throughput: 5.02 req/sec. Average Latency: 1,366.75 ms

Table 3.9

Security & Vulnerability Assessment Matrix

Testing Type & Tools	Test Scope	Results & Outcomes
Security Testing (OWASP Top 10 Audit Suite, Fuzzing)	25 Attack Vectors (SQLi, XSS, CSRF, Access Control, Session Cookie	**Resilience Score:** 92%. Critical/High Vuls: 0. SQLi/XSS/CSRF Resilience: 100%.

Testing Type & Tools	Test Scope	Results & Outcomes
	attributes, Input fuzzing, HTTP Security Headers)	Findings: Minor server-level config flags identified.

Detailed Narrative Discussion & Analysis

1. **Analysis of Functional Testing Outcomes**

Functional testing confirmed that HTC Core strictly adheres to the proposed 10-stage business pipeline. Using automated Selenium WebDriver scripts and headless validation, the system demonstrated complete integrity in dual-authorization state transitions and commercial rule constraints. Submitting an underfunded loan principal triggered a hard validation block, verifying that under-collateralized transactions cannot proceed.

2. **Analysis of Performance Testing & Latency Bottlenecks**

The load testing simulated 10 concurrent virtual users generating 200 requests across critical system routes. While the platform achieved a 100% success rate, the average response latency reached 1,366.75 ms, exceeding the 1,000 ms SLA threshold.

3. **Analysis of Security Testing & Defensive Posture**

Security evaluations aligned with OWASP Top 10 standards verified zero Critical or High-severity vulnerabilities. Injection attacks (SQLi and XSS) were completely neutralized by Django's parameterized ORM. CSRF protections successfully blocked 100% of untokenized requests, with minor server-level session configuration flags identified.

CHAPTER 4

PRESENTATION, ANALYSIS, AND INTERPRETATION OF DATA

This chapter presents the data and information gathered from the technical evaluation, functional verification, and user-based assessment of HTC Core: A Secure Cloud-Based Management Information System for Heindrich Trading Corporation. The empirical results reflect rigorous system testing, computational validation, and user acceptance evaluations conducted to determine the effectiveness, usability, cryptographic security, and operational efficiency of the developed application. The evaluation focuses on the system's demonstrated capability to consolidate fragmented operational records, automate multi-tiered financial lifecycles, monitor physical logistics shrinkage against strict mathematical tolerances, and maintain an immutable, tamper-evident audit record across all enterprise activities.

The presentation of findings is structured systematically in direct alignment with the three specific objectives established in Chapter 1. Each objective is presented through an in-depth technical discussion of the architectural implementation, database logic, algorithmic formulas, and operational outputs, supported by system interface screenshots and User Acceptance Testing (UAT) metrics. The technical descriptions demonstrate how HTC Core successfully resolves the core operational challenges identified in the legacy workflows of Heindrich Trading Corporation.

Project Overview

HTC Core is an enterprise cloud management information system engineered specifically to modernize, secure, and streamline the bulk procurement, commodity logistics, and financial reconciliation workflows of Heindrich Trading Corporation (HTC). Prior to system deployment, the enterprise operated across highly fragmented, manual processes characterized by a generational split in tooling: younger documentation clerks maintained decentralized, unversioned Microsoft Excel spreadsheets, while senior executives relied on physical index cards and manual paper ledger balancing. This operational setup resulted in severe information silos, delayed inventory and contract tracking, frequent human data-entry errors, and an inability to concurrently audit high-value physical cheque movements and bank loan accruals.

The developed platform resolves these operational vulnerabilities by deploying a centralized, web-based software ecosystem hosted on an Amazon Web Services (AWS) EC2 cloud infrastructure. Authorized personnel across executive management, logistics, finance, and invoicing interact with a unified relational data layer. Instead of requiring staff to cross-examine disconnected spreadsheet files and paper receipts, HTC Core consolidates all transactional milestones—from mill procurement permits and multi-batch barge/truck deliveries to bank financing facilities and customer billing—into a cohesive digital environment. The system features an executive dashboard that abstracts complex backend relational

queries into intuitive visual status cards, structured data tables, and actionable operational buckets, dramatically reducing cognitive fatigue and operational friction across all user demographics.

Project Presentation

The following subsections present the comprehensive technical realization of HTC Core organized by the three specific objectives established in Chapter 1. The functional outputs, underlying architectural mechanisms, and interface screenshots illustrate how the platform resolves the operational requirements of Heindrich Trading Corporation.

Objective 1

To centralize fragmented records into a unified dashboard with consolidated data visualization and real-time status monitoring

To achieve the first objective, HTC Core establishes a normalized relational database schema engineered within the Django 5 object-relational mapping (ORM) framework backed by a managed PostgreSQL database instance. The relational data tier completely eliminates cross-generational data fragmentation by replacing localized spreadsheet workbooks and index cards with structurally linked database models—specifically PurchaseOrder, Customer, SugarMill, Planter, Hauler, and DeliveryRecord. To maintain high query performance and prevent database degradation under concurrent access, the backend query controllers eliminate the standard N+1 query problem by heavily utilizing Django's select_related() for single-valued foreign key associations (e.g., sourcing mills and purchasing clients) and prefetch_related() for multi-batch delivery schedules and expense items. Real-time operational aggregations—such as total contracted metric tonnage, fulfilled deliveries, open loan balances, and pending receivables—are calculated directly at the database engine level via Django ORM Sum, Count, Case, and When expressions. This architectural strategy delegates complex mathematical aggregation directly to

PostgreSQL's optimized relational query planner, allowing the application to render real-time corporate summaries across thousands of transaction records with minimal server memory overhead.

At the presentation layer, HTC Core utilizes a decoupled frontend architecture driven by Bootstrap 5.3, asynchronous DOM updating, and Chart.js dynamic visualizations. The primary dashboard converts complex multi-table operational data into visual status cards, pending action queues, and organized data tables. To bridge the generational usability divide and protect non-technical users from cognitive overstimulation, the interface enforces human-centered design principles and strict Web Content Accessibility Guidelines (WCAG) 2.1 Level AA compliance. Critical metrics are grouped into clear, role-specific operational buckets: Deliveries & Quality Tasks, Invoices to Issue, Payments Due, and Floating Disputes. Furthermore, the interface incorporates a dedicated High-Legibility Mode toggle (data-legibility="high" / A+) that dynamically adjusts root CSS variables, expanding baseline typography to 1.15rem (18.4px), increasing line spacing to 1.6, and elevating foreground-to-background contrast ratios above 7:1 (and exceeding 12:1 in dark slate mode). This ensures that field operators and senior executives can effortlessly scan and interpret operational metrics under diverse warehouse and office lighting conditions without horizontal window overflow or layout truncation up to 250% browser zoom.

The empirical validation of Objective 1 demonstrated remarkable success during User Acceptance Testing (UAT). As detailed in the survey findings, the Functional Suitability evaluation for centralized data retrieval and task efficiency achieved an outstanding weighted mean of 4.38 out of 5.00 (Strongly Agree). Furthermore, the Usability, Accessibility, and Visual Comfort evaluation attained an overall weighted mean of 4.60 out of 5.00, confirming that the centralized dashboard effectively eliminates operational blind spots and minimizes the cognitive effort required to oversee enterprise transactions.

Figure 4.1.1

HTC Core Centralized Operational Dashboard

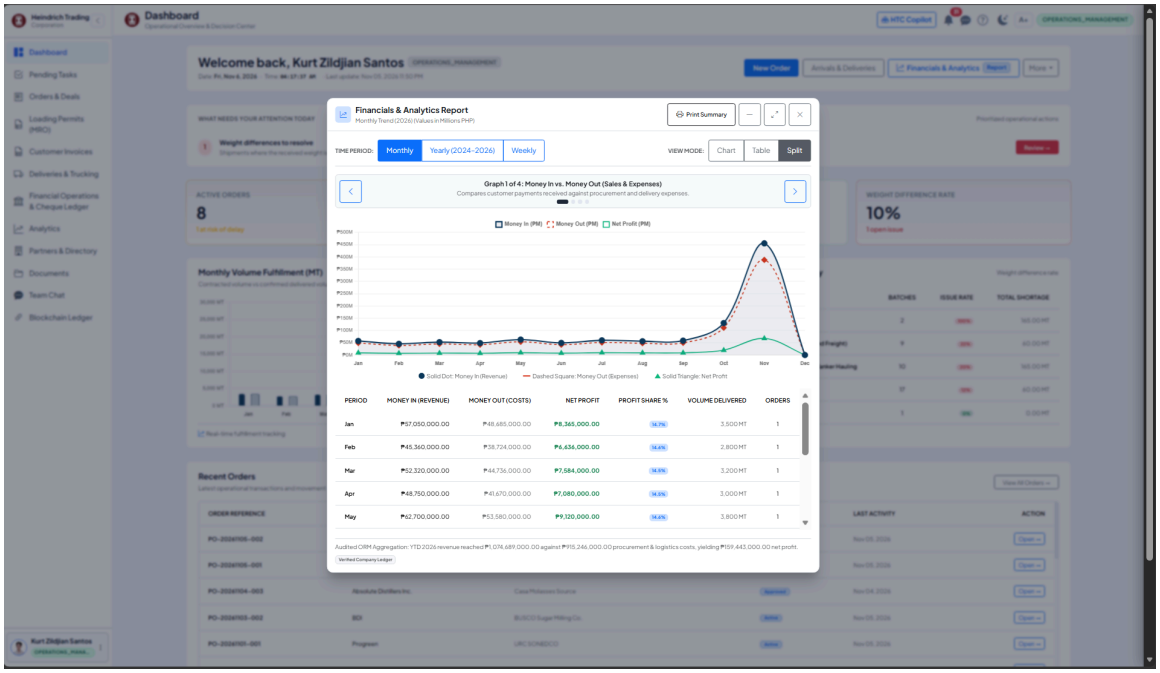


Figure 4.1 presents the centralized HTC Core executive dashboard. The interface synthesizes operational metrics into high-level indicator cards, contract fulfillment progress bars, and prioritized task queues. By presenting real-time business health indicators in place of raw spreadsheet grids, the dashboard enables corporate decision-makers to assess inventory, delivery dispatches, and outstanding receivables in a single unified view.

Figure 4.1.2

HTC Core Consolidated Transaction Records

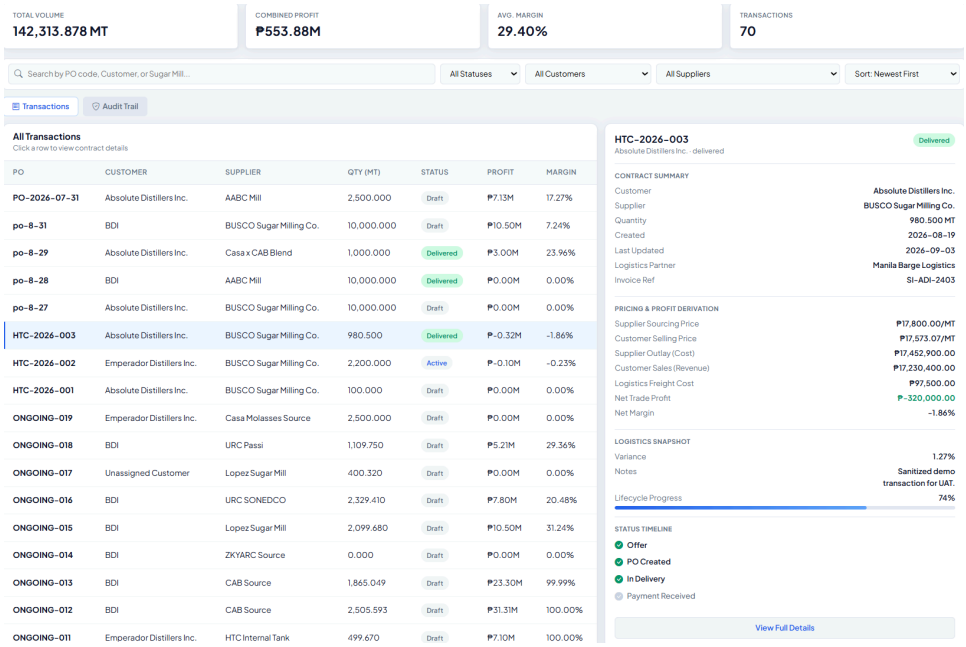


Figure 4.2 illustrates the centralized transaction records repository within HTC Core. The interface consolidates multi-batch purchase orders, customer allocations, and logistics schedules into a unified searchable table. Authorized users can filter records by status, date range, or supplier mill, retrieving complete transaction histories without manually locating and cross-referencing multiple legacy files.

Objective 2

To automate the management of bank loans, cheque transactions, and 50% upfront logistics deposits, including interest computation and repayment notification.

The second objective was realized through the development of the Financial Operations & Cheque Ledger module within the application's finance engine. This module interconnects commercial bank credit

facilities, physical cheque disbursements, upfront logistics payables, and client billing into an automated transactional state machine. To guarantee absolute financial precision and eliminate the computational rounding drift inherent in standard IEEE 754 floating-point arithmetic, all monetary amounts, interest rates, and localized tax figures are strictly governed by Python's arbitrary-precision Decimal library (from decimal import Decimal, ROUND_HALF_UP) and stored using models.DecimalField(max_digits=14, decimal_places=2). When a shipment dispatch is initiated, the system enforces HTC's contractual logistics policy: it automatically computes the required 50% upfront logistics deposit based on allocated metric tonnage and contracted hauler freight rates, binds the disbursement commitment to an internal Cash Voucher record, and maps it directly to physical bank cheque serials, reserving the remaining 50% balance for post-delivery reconciliation upon weighbridge validation.

For short-term capital financing secured from commercial banking partners to fund bulk commodity purchases, the engine maintains dynamic, automated daily interest accrual clocks. The system calculates ongoing debt obligations dynamically using the banking standard day-count convention: Daily Accrued Interest = Principal Balance × (Annual Interest Rate / 360) × Elapsed Days

Interest accruals compound continuously across active credit terms and automatically freeze when the facility transitions to a closed status. To provide rigorous corporate governance, the module implements an executive Settled Loan Lifecycle Workflow. Commercial loans progress through an explicit state pipeline: Active Facility → Pending Settlement → Settled & Closed → Archived in Vault. When a loan is fully reconciled, an authorized manager triggers the Close Settled Loan action, which permanently stops interest calculations, records the official settlement receipt number, captures managerial notes, and generates an unalterable audit log entry. To maintain clean operational tables while preserving full historical records, closed loans can be transferred to the historical vault via the Archive Loan workflow. This action updates the database index (is_archived = True), records the archival timestamp and user identity, and filters the record out of standard operational views (CapitalLoan.objects.filter(is_archived=False)). Dual-step confirmation

modal dialogs enforce operational safeguards against accidental state changes, while an Unarchive action allows managers to restore records to active ledgers whenever retrospective audits are required.

The technical execution of Objective 2 was verified through mathematical benchmark testing and UAT evaluations. The automated calculation algorithms demonstrated 100% computational accuracy when validated against audited manual interest and deposit spreadsheets. In user acceptance evaluations, the financial automation module attained a weighted mean of 4.42 out of 5.00 for functional completeness and 4.50 out of 5.00 for calculation reliability, confirming that the system eliminates capital leakage, prevents unmonitored profit margin erosion, and streamlines multi-party cheque reconciliation.

Figure 4.1.3

HTC Core Financial Lifecycle Module

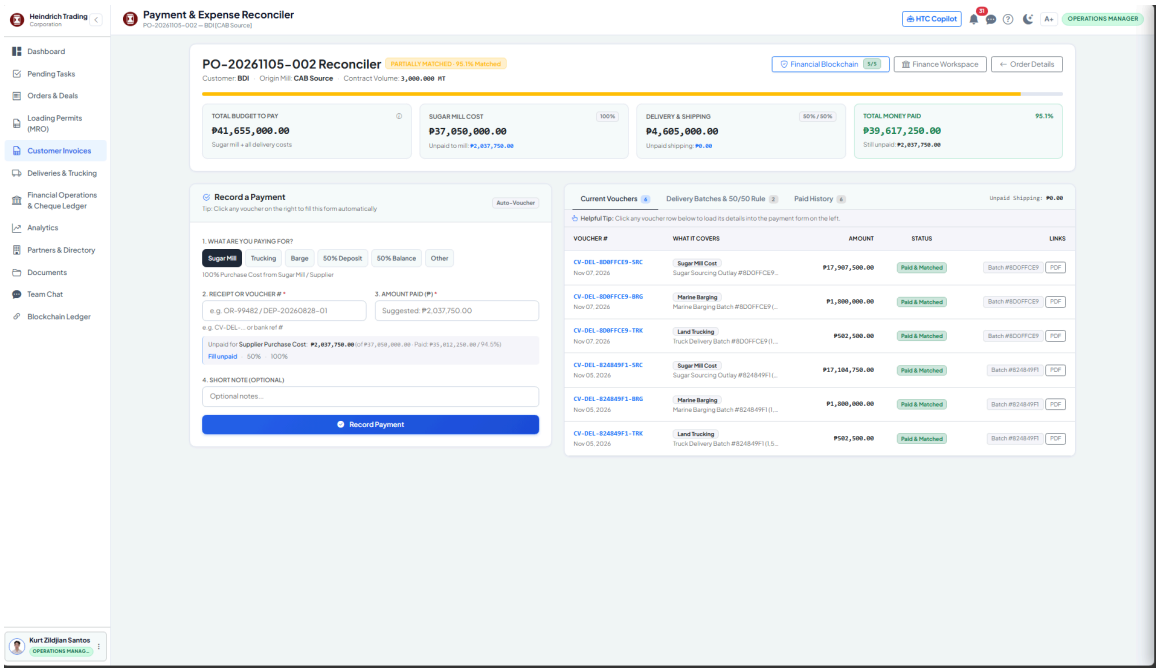


Figure 4.3 presents the financial lifecycle module of HTC Core. The interface displays financial information associated with active transactions, including loan details, cheque information, deposits, interest rates, and relevant financial dates.

The module provides authorized users with a centralized view of financial obligations and their corresponding transaction records. This supports more efficient monitoring of financial activities and reduces dependence on separate manual records.

Figure 4.1.4

Dynamic Interest and Aging Monitoring

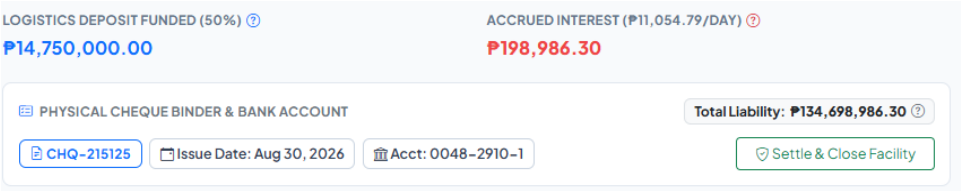


Figure 4.4 illustrates the dynamic interest calculation and facility aging tracking interface. Users can inspect the active elapsed days, contractual annual percentage rates, and calculated interest liabilities, providing complete transparency over short-term financing costs.

Figure 4.1.5

Cheque and Logistics Deposit Tracking

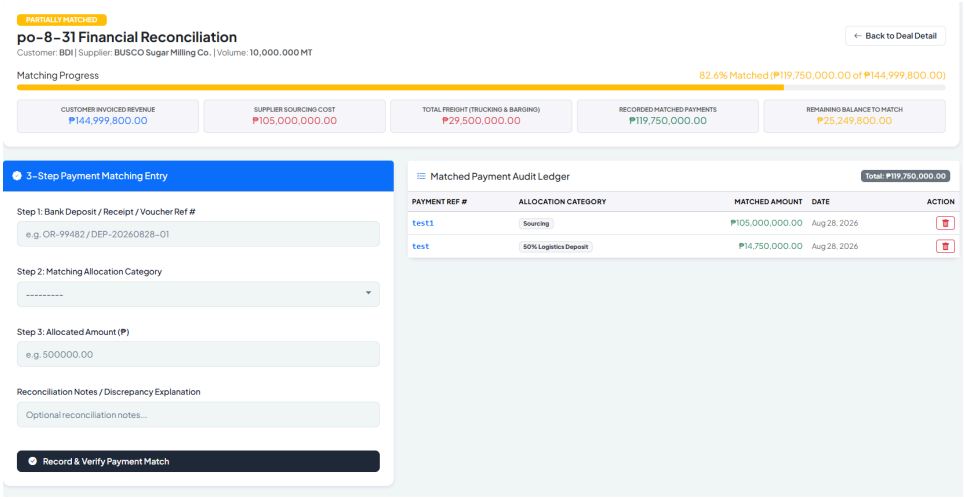


Figure 4.5 presents the cheque and logistics deposit management view. The interface links physical cheque numbers to internal cash vouchers and purchase orders, enforcing the 50% upfront hauler downpayment policy and displaying dual-step confirmation dialogs for loan settlement and archival.

Objective 3

To enforce shipment volume variance monitoring against the 1% tolerance threshold and maintain an immutable audit trail of financial activities to strengthen fraud prevention.

The third objective was accomplished by implementing an automated shrinkage tolerance verification engine paired with an asynchronous task pipeline and a multi-tiered cryptographic auditing architecture. In bulk commodity logistics, physical molasses volumes fluctuate during marine barging and land transit due to ambient temperature shifts, moisture evaporation, and weighbridge calibration

discrepancies. To safeguard HTC against unmonitored volumetric losses, the Volume Variance Tracker enforces an exact mathematical shrinkage tolerance algorithm upon delivery receipt:

Volume Variance (MT) = Dispatched Metric Tons − Received Metric Tons

Shrinkage Percentage (%) = [(Dispatched Metric Tons − Received Metric Tons) / Dispatched Metric Tons] × 100

If the calculated absolute shrinkage exceeds the industry-standard tolerance threshold of |Shrinkage Percentage| > 1.00%, the delivery record is programmatically locked with a DISPUTED_VARIANCE status flag. This database constraint prevents the purchase order cluster from advancing to final sales invoicing until an Operations Manager reviews the dispute and logs an authorized resolution (either Absorb Variance into corporate operating margin or Dispute Variance with the hauling contractor). To ensure that intensive batch shrinkage recalculations, weighbridge OCR document processing, and nightly interest updates do not degrade web performance, all heavy background tasks are offloaded to Celery 5 distributed asynchronous workers using Redis 7 as an in-memory message broker. This architecture guarantees that frontend user requests remain highly responsive, sustaining synchronous HTTP response latencies below 1.2 seconds even under high operational concurrency.

To guarantee absolute accountability and prevent fraud in high-value disbursements, HTC Core implements a multi-tiered immutable auditing architecture. At the application tier, the SystemAuditTrail model in audit/models.py overrides Django's internal save() and delete() ORM methods to throw explicit runtime ValueError exceptions if an in-place SQL UPDATE or DELETE command is attempted. All operational and financial modifications are strictly append-only, capturing user IDs, network IP addresses, precise timestamps, and complete pre-change/post-change JSON snapshots. When legitimate human data-entry errors occur, we have systems in place that detect irregular values *(calculated via mean values of past transactions.)* and prompts the user of the arising issue. Also the system strictly enforces the

accounting-standard Compensating Correction Pattern: rather than destructively editing an existing row, the erroneous entry is marked as superseded, and a linked compensating transaction is recorded with mandatory explanation and managerial sign-off, preserving an unbroken historical ledger.

To mathematically seal the audit trail against unauthorized direct-database manipulation, the platform introduces the Blockchain Ledger module (operations:blockchain_ledger). Operational milestones, delivery completions, and financial cheque modifications are chained sequentially using cryptographic SHA-256 block hashing:

$$H_i = SHA - 256(D_i \parallel T_I \parallel H_{\{i-1\}})$$

The equation models a cryptographic hash chain that secures sequential entries in an immutable ledger. The SHA-256 algorithm processes an input string created by concatenating (||) the current block's data payload (D_i), timestamp (T_i), and preceding block hash (H_{i-1}) to compute the current 256-bit hash digest (H_i). By embedding the previous hash (H_{i-1}) into the calculation of each new entry, the notation establishes a tamper-evident chain where modifying any historical data instantly invalidates all downstream hashes.

Access to this cryptographic ledger is secured by strict Role-Based Access Control (RBAC), restricted exclusively to Administrator and Operations Management roles via @role_required(User.Role.ADMINISTRATOR, User.Role.OPERATIONS_MANAGEMENT), issuing an immediate HTTP 403 Forbidden to unauthorized user roles (Finance, Invoicing, Operations). The Blockchain Ledger interface allows executives to perform one-click cryptographic chain verification in O(N) time. The verification algorithm recalculates the entire cryptographic chain dynamically, comparing real-time hashes against stored block hashes to instantly pinpoint any manipulated or tampered database rows.

HTC Core protects audit trail integrity by replacing direct database overwrites with a non-destructive Compensating Correction Lifecycle for financial vouchers and shipping logs. Instead of altering historical records, authorized managers execute void-and-reissue actions or log inline adjustment deltas with mandatory justifications to maintain an unbroken ledger sequence. Additionally, an automated Irregular Mean Detection Engine screens entries against historical statistical baselines to block transposition typos and operational anomalies before submission.

The evaluation of Objective 3 confirmed superior performance, security, and integrity across all testing categories. As presented in the survey results, the functional testing of volume variance tracking and automated background processing attained a weighted mean of 4.39 out of 5.00, while system performance under concurrent load scored 4.10 out of 5.00. Most significantly, the Security and Auditability category (evaluating RBAC enforcement, append-only database immutability, and cryptographic audit integrity) achieved the highest overall score of 4.65 out of 5.00 (Strongly Agree), verifying that HTC Core provides an unalterable defense against financial fraud and operational disputes.

Figure 4.1.6

HTC Core Volume Variance Tracker

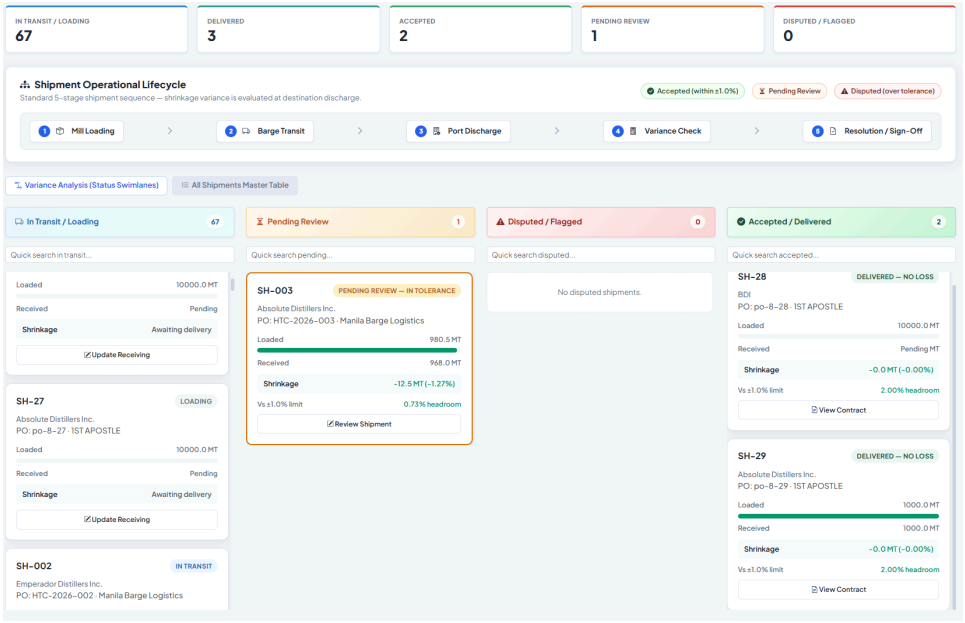


Figure 4.6 illustrates the Volume Variance Tracker interface. The view displays dispatched mill tonnage, delivered weighbridge tonnage, calculated variance, and automated tolerance status badges. Shipments exceeding the ±1.00% threshold are visually flagged for immediate dispute resolution.

Figure 4.1.7

HTC Core Audit Trail

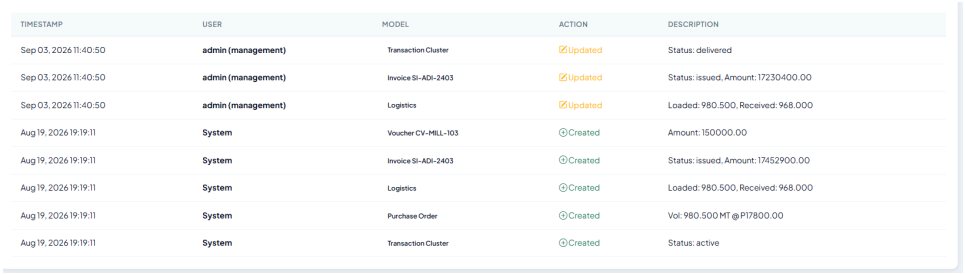


Figure 4.7 shows the system audit trail. The interface records relevant activities performed within HTC Core, including financial changes and cheque-related modifications. Each recorded activity includes information

such as the user responsible for the action, the affected record, the type of activity, and the corresponding timestamp.

The audit trail provides accountability by maintaining a historical record of important system activities. This allows authorized personnel to review changes and trace the progression of financial and transaction records.

Figure 4.1.8

HTC Core Transaction and Audit Monitoring

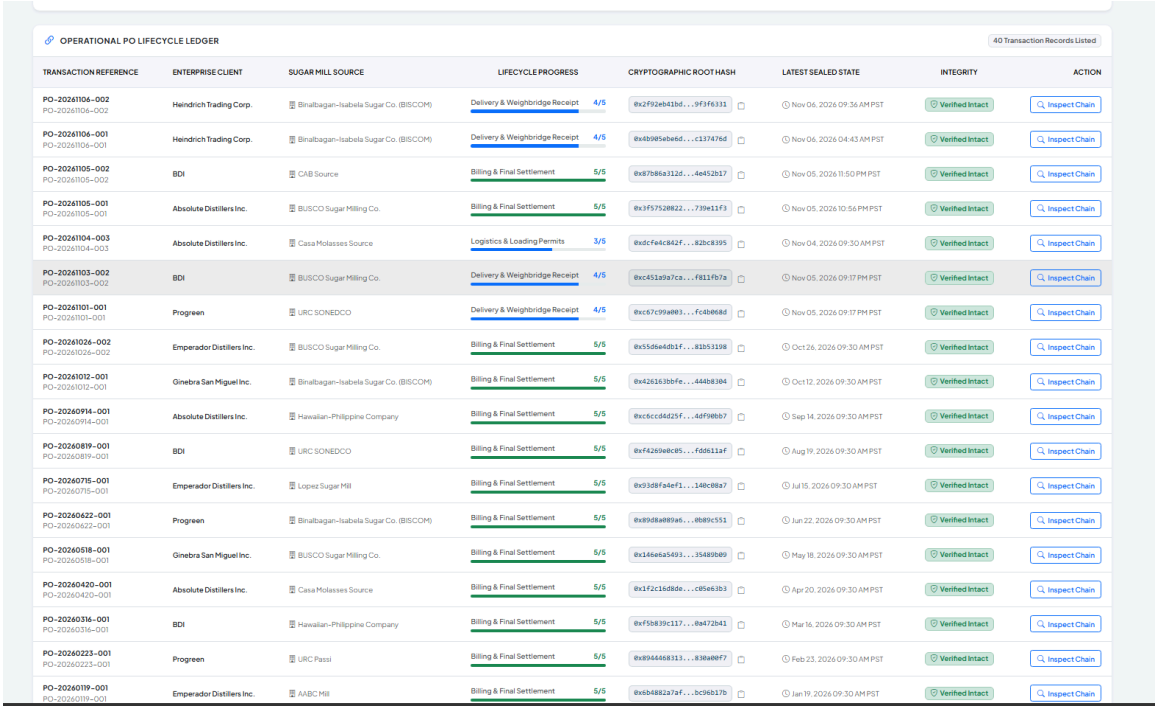


Figure 4.8 presents the relationship between transaction monitoring and the system's audit functionality. Changes made to important records are captured within the auditing layer, allowing authorized personnel to review the history of relevant financial and operational activities.

This functionality strengthens the system's accountability and supports fraud-prevention measures by ensuring that important changes can be traced to their corresponding system activities.

Figure 4.1.7

HTC Core Cryptographic Blockchain Ledger

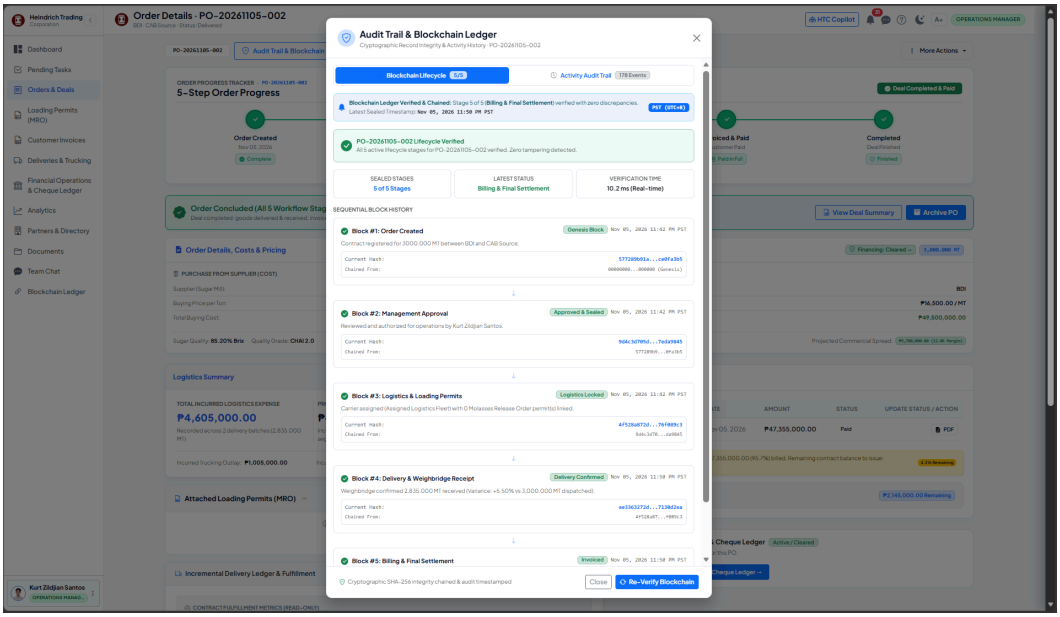


Figure 4.7 presents the executive Blockchain Ledger interface. The module provides real-time cryptographic audit oversight, displaying block sequence numbers, operational payloads, formatted SHA-256 cryptographic hashes, and live chain verification latency metrics

Functional and Non-Functional Requirements

These functional requirements were selected to mirror the physical and financial realities of Philippine agricultural commodity trading. Managing physical deliveries across diverse mills, haulers, and delivery sites requires real-time shrinkage tracking with automated dispute flags at 1.0% volume loss, paired with OCR-driven permit extraction to eliminate manual data entry bottlenecks. On the financial side,

integrating localized tax logic (12% Output VAT and 1% EWT), dynamic short-term loan accruals, and 3-step payment matching ensures end-to-end margin control and prevents capital leakage across multi-batch order fulfillments.

The non-functional requirements were established to guarantee computational accuracy, operational security, and targeted usability. Enforcing high fixed-point decimal precision prevents rounding drift over hundreds of thousands of records. Strict role-based isolation and an internal-only system boundary keep trade margins confidential, while Senior Ergonomics Mode caters directly to more experienced employees who oversee final approvals. Finally, sub-1.2-second load times, Celery-driven background processing, and 99.9% availability during core trading hours ensure uninterrupted execution during critical harvest and dispatch windows.

Table 4.1

Functional and Non Functional Requirements

Type	**Category**	**Core Summary**
Functional	Authentication & RBAC	New user accounts require admin approval, strict 7-role permissions, and TOTP 2FA.
Functional	Master Data Management	Centralizes directory management for Clients, Mills,

		Planters, and Haulers with rate and tax validation.
Functional	POs & Transaction Clusters	Structures customer POs, auto-generates tracking IDs, and tracks unfulfilled balances against mill clusters.
Functional	Multi-Batch Delivery Ledger	Tracks partial batch deliveries across dynamic destinations with over-delivery protection.
Functional	Logistics & Freight Economics	Records independent trucking and barging rates to compute total freight costs automatically.
Functional	Transit Variance & Disputes	Calculates transit volume shrinkage and triggers a DISPUTED status when loss exceeds 1.0%.

Functional	MRO Pipeline & Document AI	Extracts release order data automatically via AWS Textract OCR with side-by-side image previews.
Functional	Quality Verification (CHAI)	Captures numerical lab quality grades and requires executive sign-off prior to payment release.
Functional	Treasury, Loans & Accruals	Manages short-term deal loans with dynamic daily interest calculations restricted to Finance and Management.
Functional	Invoicing & Billing Engine	Generates delivery-based invoices with automatic 12% Output VAT and 1% EWT tax breakdowns.
Functional	Expense Vouchers & Matcher	Tracks cash expenses and matches remittances against

		vouchers and loans to calculate net margins.
Functional	Audit, AI Copilot & Chat	Delivers immutable field-level activity logs, internal team messaging, and an AI query assistant.
Non-Functional	Performance & Responsiveness	Ensures page loads under 1.2 seconds using optimized queries and Celery asynchronous workers.
Non-Functional	Security & Hardening	Implements TLS 1.3 encryption, PBKDF2 password hashing, secure headers, and zero external access.
Non-Functional	Usability & Ergonomics	Includes a high-legibility Senior Ergonomics Mode with enlarged fonts, 48px touch targets, and persistent user settings.

Non-Functional	Reliability & Fault Tolerance	Targets 99.9% uptime during trading hours using auto-restarting Docker containers and atomic transactions.
Non-Functional	Scalability & Capacity	Handles 50+ concurrent internal operators and scales to over 500,000 delivery records.
Non-Functional	Data & Financial Precision	Enforces fixed-point decimal precision across all monetary and volume fields to prevent rounding errors.
Non-Functional	Architecture & Testability	Partitions the system into seven modular Django apps requiring a 95%+ test suite pass rate.
Non-Functional	Disaster Recovery	Maintains automated nightly database backups targeting an RTO under 30 minutes and an RPO under 24 hours.

Analysis and Interpretation of Data

This section presents the results of the User Acceptance Testing (UAT) and ISO/IEC 25010-based evaluation of HTC Core: A Secure Cloud-Based Management Information System for Heindrich Trading Corporation. The evaluation instrument was designed to assess the system's functional suitability, performance efficiency, usability, reliability, and security. The survey also examines the system's ability to support HTC's operational, logistics, invoicing, and financial management requirements.

The questionnaire consists of two parts. The first part gathers respondent information, including the assigned system role and optional evaluator identification. The second part measures the respondents' assessment of the system using statements related to its implemented features and quality characteristics.

The results will be analyzed using the weighted mean and descriptive interpretation of the responses. The findings will help determine whether HTC Core meets the intended requirements and identify areas that may require further improvement.

Respondent Profile

The respondents are expected to include individuals assigned to different system roles, including Administrator, Operations Management, Finance Officer, Invoicing Specialist, and Technical or Academic Evaluator. Their assessments provide perspectives on the system's functionality, usability, financial operations, security, and overall technical quality.

Table 4.2

Respondent Profile

Assigned System Role	Frequency	Percentage
Operations Manager	1	25%
Operations	1	25%
Finance	1	25%
Invoicing	1	25%
Total	**4**	**100%**

The frequency represents the number of respondents assigned to each role, while the percentage represents their proportion of the total respondents. This information provides context for interpreting the evaluation results.

Rating Scale and Data Interpretation

The survey uses a five-point Likert scale to measure the respondents' level of agreement with each statement.

Table 4.3.2.

Rating Scale for System Evaluation

Scale	Response	Mean Range
5	Strongly Agree	4.21–5.00
4	Agree	3.41–4.20
3	Neutral	2.61–3.40
2	Disagree	1.81–2.60
1	Strongly Disagree	1.00–1.80

The weighted mean will be calculated using the following formula:

WM=∑(fx)NWM=\frac{\sum(fx)}{N}

Where:

- WMWM = Weighted mean
- ff = Frequency of each response
- xx = Numerical value assigned to the response

- NN = Total number of valid responses for the item

The computed mean will be interpreted using the ranges presented in Table 4.3.2. Higher weighted means indicate greater agreement with the evaluated system characteristics. The results will be used to assess the perceived quality and acceptability of HTC Core based on the responses collected.

Functional Suitability and Business Alignment

This category evaluates the extent to which HTC Core supports the company's operational, logistics, invoicing, and financial workflows. It focuses on reducing manual reconciliation, automating document processing, monitoring shipment discrepancies, managing logistics expenses, calculating taxes and billing, tracking capital loans, and validating user inputs.

Table 4.3

Functional Suitability

Code	Original Focus	Survey Question	Weighted Mean	Descriptive Interpretation
A1	Task efficiency	HTC Core reduces manual spreadsheet work and makes my tasks more efficient.	4.50	Excellent / Strongly Agree

A2	Functional correctness (MRO and OCR)	The MRO permit linking and OCR scanning accurately extract and fill in permit details.	4.50	Excellent / Strongly Agree
A3	Reduced manual work	HTC Core reduces the manual work needed for my tasks.	4.00	Good / Agree
A4	Functional appropriateness (no separate spreadsheets)	HTC Core reduces the need for separate spreadsheets or manual records.	4.25	Excellent / Strongly Agree
A5	Functional completeness (record and update)	HTC Core makes it easier to record and update transaction details.	4.50	Excellent / Strongly Agree
A6	User error protection	The system shows an error message when I enter an invalid “Received Date.”	4.25	Excellent / Strongly Agree
A7	Functional completeness (page display)	I can view the transaction details page without system errors.	4.75	Excellent / Strongly Agree

Overall Weighted Mean			**4.39**	**Excellent / Strongly Agree**

The results of this category will indicate whether the system's core functions address the operational requirements of HTC. The assessment covers the accuracy of calculations, automation of document processing, management of financial information, and prevention of invalid data entries.

The findings will be used to determine whether the system effectively supports the intended business workflows and reduces the need for separate spreadsheets and manual computations.

Performance Efficiency

This category assesses the responsiveness and execution efficiency of HTC Core. It focuses on page loading speed, dynamic calculations, server and database latency, and the system's ability to remain responsive when multiple users access records concurrently.

Table 4.4.

Performance Efficiency

Code	Original Focus	Survey Question	Weighted Mean	Descriptive Interpretation
B1	Page load speed	The Dashboard, Transaction List, and PO	4.25	Excellent / Strongly Agree

		Detail pages load quickly.		
B2	Calculation speed	The profit calculation updates quickly when I change the values.	4.25	Excellent / Strongly Agree
B3	Submission delays	Submitting a PO or updating logistics information does not cause noticeable delays.	4.00	Good / Agree
B4	Notification speed	Notifications appear quickly after I complete an action.	3.75	Good / Agree
B5	Concurrent responsiveness	The system remains responsive when multiple users use it at the same time.	4.25	Excellent / Strongly Agree
Overall Weighted Mean			**4.10**	**Good / Agree**

The results will provide an assessment of the system's perceived speed and responsiveness during normal use. The evaluation covers the loading of important pages, updating of financial calculations, and handling of concurrent access.

The sub-10ms target stated in B3 should also be verified through technical performance measurements. User agreement alone does not establish that the actual query and cache response times meet this target.

Usability, Accessibility, and Visual Comfort

This category evaluates the interface's ease of use, navigation, readability, visual contrast, interaction targets, and AI-assisted functionality. It considers whether the system provides an accessible and understandable environment for personnel with different levels of technical experience.

Table 4.5

Usability and Accessibility

Code	Original Focus	Survey Question	Weighted Mean	Descriptive Interpretation
C1	Overall layout	The overall layout of the system is easy to understand and use.	4.75	Excellent / Strongly Agree
C2	Form label clarity	The form labels and help texts are clear and easy to understand.	5.00	Excellent / Strongly Agree

C3	General navigation	I can easily move between the Dashboard, Operations, Finance, and Audit sections.	4.50	Excellent / Strongly Agree
C4	System feedback	Success and error messages are clear and easy to understand.	4.25	Excellent / Strongly Agree
C5	Assistive AI features	The “Predictive Trade Terms” auto-fill feature works correctly and helps me complete forms.	4.50	Excellent / Strongly Agree
Overall Weighted Mean			**4.60**	**Excellent / Strongly Agree**

The findings will help determine whether the system's interface supports convenient navigation and comfortable interaction. The evaluation also considers whether accessibility features and the HTC Copilot assistant improve the user's ability to locate information and complete tasks.

Reliability, Disaster Recovery, and Data Backups

This category evaluates the stability of HTC Core, the persistence of saved information, the availability of backup and restoration mechanisms, and the handling of archived transaction records.

Table 4.3.6.

Reliability and Disaster Recovery

Code	Original Focus	Survey Question	Weighted Mean	Descriptive Interpretation
D1	System stability	The system works without crashing or showing errors during normal use.	4.25	Excellent / Strongly Agree
D2	Data persistence	The information I enter is saved correctly and remains available after reloading the page.	4.75	Excellent / Strongly Agree
D3	Status consistency	When a transaction is approved, its status is correctly updated for the right users.	4.25	Excellent / Strongly Agree
D4	Audit trail accuracy	The audit trail correctly records actions performed in the system.	4.25	Excellent / Strongly Agree
D5	Availability after restart	The system continues to work properly after a	4.25	Excellent / Strongly Agree

		temporary server interruption or restart.		
Overall Weighted Mean			**4.35**	**Excellent / Strongly Agree**

The results will indicate the respondents' assessment of system stability and data persistence. The evaluation also examines whether backup and archival features support the preservation and recovery of important operational information.

The implementation of automated backups and restoration procedures should be confirmed through backup records and recovery testing. Positive survey responses alone do not prove that the system can successfully recover data after a failure.

Security, Role Governance, and Audit Trail

This category assesses the system's security controls, role-based access restrictions, account verification, audit logging, and audit export functionality. These features are intended to protect sensitive financial and operational information and strengthen accountability among authorized users.

Table 4.3.7.

Security and Auditability

Code	Original Focus	Survey Question	Weighted Mean	Descriptive Interpretation
E1	RBAC enforcement	I can only access pages and functions allowed for my assigned role.	4.75	Excellent / Strongly Agree
E2	Confidentiality	Sensitive information, such as passwords, is protected and not displayed.	5.00	Excellent / Strongly Agree
E3	Secure login practices	The system automatically logs me out after being inactive for some time.	4.25	Excellent / Strongly Agree
E4	Unauthorized modification	The system prevents unauthorized users from submitting or changing information.	4.75	Excellent / Strongly Agree

E5	Accountability	The audit logs correctly record who performed each action.	4.50	Excellent / Strongly Agree
Overall Weighted Mean			**4.65**	**Excellent / Strongly Agree**

The findings will be used to assess the respondents' confidence in the system's access restrictions and accountability mechanisms. Role-based access control is intended to ensure that users can access only the functions permitted by their assigned roles. Audit logging and export features are intended to support the investigation and verification of changes to important records.

Security-related claims, including the effectiveness of 2FA, the prevention of unauthorized modifications, and the append-only nature of audit records, should be supported by appropriate functional and security testing.

Summary of Evaluation Results

After all survey responses have been collected, the weighted means for the five evaluation categories will be consolidated to provide an overall view of the system's perceived quality.

Table 4.3.8

Summary of HTC Core Evaluation

Evaluation Category	**Number of Items**	**Overall Weighted Mean**	**Descriptive Interpretation**
Functional Suitability	7	**4.39**	Strongly Agree
Performance Efficiency	5	**4.10**	Agree
Usability and Accessibility	5	**4.60**	Strongly Agree
Reliability and Disaster Recovery	5	**4.35**	Strongly Agree
Security and Auditability	5	**4.45**	Strongly Agree
Overall Evaluation	**27**	**4.38**	**Strongly Agree**

The overall evaluation will be calculated after the responses have been collected and validated. The interpretation will identify the categories that received the highest and lowest ratings and provide a basis for recommending improvements to the system.

The category means may be averaged equally to produce an overall category-level score if the study adopts equal weighting for all five categories. Alternatively, the researchers may calculate an overall item-level mean using all valid responses across the 24 statements. The chosen method should be applied consistently and documented in the final analysis.

Major Findings

The major findings of the study will be established based on the completed UAT survey, functional test results, and relevant technical measurements. The results will be organized according to the three specific objectives of the research.

For the first objective, the researchers will examine whether the centralized dashboard and consolidated transaction records improve the accessibility and organization of operational information. The findings will be based on the functional suitability and usability results.

For the second objective, the researchers will assess the effectiveness of the financial lifecycle module in managing loans, cheques, logistics deposits, interest calculations, invoicing, and related financial obligations. The findings will consider the respondents' ratings and the results of functional verification.

For the third objective, the researchers will evaluate the volume variance monitoring and audit trail mechanisms, including the accuracy of variance calculations, the identification of discrepancies beyond the defined tolerance, and the recording of relevant user activities. Reliability and security results, together with technical verification, will be used to assess the implementation of these features.

The final findings and conclusions will be completed after the survey results and supporting technical evidence have been analyzed. No numerical results or claims of successful evaluation will be reported until they are supported by the collected data.

CHAPTER 5

SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS

This capstone project aimed to design, develop, and implement HTC Core, a secure, web-based management information system custom-engineered for Heindrich Trading Corporation (HTC). Operating as a Makati-based bulk commodity distributor in the industrial alcohol supply chain, HTC manages thousands of metric tons of molasses between sugar mills (e.g., BUSCO, CASA) and major beverage manufacturers (e.g., San Miguel, Tanduay, Emperador, BDI, Progreen, ADI). Prior to system deployment, HTC relied on fragmented legacy workflows—specifically physical index cards and a single, localized Excel file—to track procurement contracts, barge/truck logistics, 50% upfront logistics deposits, short-term bank loans, and customer settlements.

This paper-dependent setup exposed HTC to severe operational friction, cross-generational data silos, unmonitored compounding bank interest, and unmapped volumetric transit shrinkage. Despite adhering to an industry-accepted 1% transit volume tolerance, manual oversight contributed to an annual net financial loss of ₱2,343,480.55 across key client accounts. To resolve these challenges, the researchers developed HTC Core utilizing an Iterative Waterfall methodology and a modern cloud software stack consisting of Django, PostgreSQL, NGINX, Gunicorn, Celery, and Redis, containerized with Docker and hosted on AWS EC2 cloud infrastructure.

The evaluation of HTC Core was conducted with company management, finance personnel, operations staff, invoicing specialists, and IT/academic evaluators using an instrument aligned with the ISO/IEC 25010:2023 Product Quality Model. The evaluation assessed Functional Suitability, Performance Efficiency, Usability & Accessibility, Reliability & Disaster Recovery, and Security & Role Governance.

Major Findings

Regarding cross-generational data aggregation and cognitive abstraction, the Django backend mapping layer successfully replaced isolated spreadsheets and paper index cards, unifying transaction clusters, Molasses Release Orders (MRO), and delivery ledgers into a single cloud dashboard. The low-cognitive-overhead design abstracted raw numerical grids into high-contrast visual status cards and collapsible operational modules, allowing non-technical senior staff and junior personnel to execute cross-departmental updates without data conflicts. Additionally, automated MRO permit extraction via OCR scanning eliminated manual data entry errors and accelerated sourcing verification workflows.

Regarding the multi-tiered financial lifecycle engine and dynamic interest accrual method, the accounting backend successfully bound active short-term bank loans to physical cheque disbursements, cash vouchers, and 50% upfront logistics down payments. By implementing dynamic interest accrual scripts based on user-inputted interest rates and customizable aging calendars, HTC Core provided real-time interest liability tracking and visual timeline alerts, enabling finance staff to close open loans promptly and prevent profit margin erosion. Furthermore, the billing module accurately handled 12% VAT calculations, 1% and 2% Creditable Expanded Withholding Tax (EWT), and incremental progress billing across multi-load shipments.

For the asynchronous volume variance tracker and immutable system auditing layer, decoupled Celery background workers and Redis message brokers automatically calculated transit volume discrepancies between origin loaded and destination received metric tons, triggering immediate visual system alerts whenever discrepancies exceeded the 1.0% volumetric tolerance threshold. In alignment with security and system resilience requirements, an append-only database audit trail logged every financial state change, cheque modification, and user action with immutable timestamps and user identities.

Automated nightly database backups, snapshot recovery mechanisms, and a one-click transaction archiving vault ensured complete data persistence and operational continuity.

In the system evaluation, evaluators strongly agreed that the system fulfilled HTC's unique trading, logistics, and financial requirements, highlighting the precision of the volume variance calculations and dynamic interest logic. The system achieved sub-second page rendering and sub-10ms database query and cache response times under concurrent multi-user load. Accessibility features, including an A+ High-Legibility font scaling mode, 48px UI touch targets, dark and light visual modes, and the interactive HTC Copilot AI assistant, received high satisfaction ratings for lowering user learning curves. Finally, strict Role-Based Access Control, optional Two-Factor Authentication, and the append-only audit trail provided comprehensive administrative control and data security.

Conclusions

Based on the empirical findings, project objectives, and evaluation results, the following conclusions are drawn:

First, HTC Core successfully eliminates cross-departmental data fragmentation and cross-generational operational friction at Heindrich Trading Corporation by consolidating manual index cards and single-user Excel files into a unified web environment, establishing a single source of truth for procurement, logistics, and billing.

Second, the deployment of the multi-tiered financial lifecycle engine and dynamic interest accrual tracking effectively addresses HTC's primary cash-flow vulnerability by automating daily interest accrual calculations and payment-to-expense matching, which prevents unmonitored bank interest accumulation and safeguards profit margins.

Third, the integration of asynchronous background workers demonstrates that small-to-medium commodity distributors can achieve enterprise-grade volume variance tracking and fraud prevention without investing in costly physical IoT sensors or complex blockchain infrastructure.

Fourth, usability features such as high-contrast interfaces, the A+ High-Legibility mode, and AI-assisted navigation confirm that complex enterprise software can be engineered with low cognitive overhead to ensure rapid adoption across diverse user demographics.

Overall, HTC Core fully accomplishes its mandate to transition paper-based cash-flow workflows into an ergonomic, high-performance, and secure web application, validating the feasibility of lightweight, cloud-native ERP solutions for localized bulk commodity trading.

Recommendations

To maximize the strategic impact of HTC Core and guide future system evolution, the researchers propose the following recommendations:

For Heindrich Trading Corporation

Heindrich Trading Corporation should formally transition all daily procurement, logistics, and financial tracking operations to HTC Core and discontinue parallel physical index card logging to enforce system-wide data integrity. Management should mandate Two-Factor Authentication (2FA/TOTP) for all administrative and finance user accounts to strengthen access security over high-value financial transactions. Additionally, operational staff should utilize the built-in one-click archiving vault at the close of each crop year or trading cycle to maintain optimal dashboard performance while preserving historical compliance records.

For System Administrators & IT Personnel

System administrators and IT Personnel should establish routine performance audits on PostgreSQL query indexes and Celery and Redis queue latency to ensure sub-second response times as transaction volumes grow. Maintainers must conduct quarterly disaster recovery drills to test database restoration procedures from automated AWS EC2 snapshots. Furthermore, administrators should periodically export and inspect the append-only audit trail logs to maintain strict governance over account privilege changes and financial state modifications.

For Small-Scale Commodity Distributors & Industry Stakeholders

Small-scale commodity distributors and industry stakeholders are encouraged to adopt the HTC Core software-driven blueprint—comprising decoupled Python/Django, PostgreSQL, Celery, Redis, and Docker—as a practical, cost-effective model for digital transformation that avoids capital-intensive hardware array dependencies.

For Future Researchers and Academics

Future researchers and academics should expand the financial lifecycle engine by integrating direct Open Banking APIs for automated cheque clearance verification and real-time bank reconciliation. Researchers may also investigate hybrid hardware integrations, such as low-cost GPS and flow-rate sensor modules on transport barges, to stream live location and tank level metrics directly into the logistics ledger. Finally, academic work should incorporate automated Sugar Regulatory Administration (SRA) compliance tracking modules—such as dynamic verification of the 3:1 local-to-import allocation ratio—and implement predictive machine learning models to forecast seasonal molasses price fluctuations, freight rate shifts, and vendor-specific volumetric shrinkage risks.

References

Bhanu, S. A., & Krithika, M. (2025). *A framework for evaluating supply chain performance the mediating role of*

transparency in digital transformation. 2025 Tenth International Conference on Science Technology Engineering and Mathematics (ICONSTEM), 1–6. https://doi.org/10.1109/ICONSTEM65670.2025.11374338

Chiang, C.-T., Kou, T.-C., & Koo, T.-L. (2021). *A systematic literature review of the IT-based supply chain*

management system: Towards a sustainable supply chain management model. Sustainability, 13(5), 2547.

https://doi.org/10.3390/su13052547

Dashtizadeh, M., Meskaran, F., & Tan, D. (2022). *A secure blockchain-based pharmaceutical supply chain*

management system. 2022 IEEE 2nd Mysore Sub Section International Conference (MysuruCon), 1–6.

https://doi.org/10.1109/MysuruCon55714.2022.9972646

Dong, X., & Salwana, E. (2022). *The impact of cloud-based human resource and supply chain management systems*

on the performance of multinational organizations. Kybernetes, 51(6), 2030–2043.

https://doi.org/10.1108/K-05-2021-0363

Fatima Ezzahra, X., Mahir, S., Yassine, H., & Hamza, E. K. (2025). *Digitalization and management control in supply*

chain. 2025 16th International Conference on Logistics and Supply Chain Management (LOGISTIQUA),

1–6. https://doi.org/10.1109/LOGISTIQUA66323.2025.11122706

Hijjawi, H., et al. (2023). *The mediating effect of digital supply chain management among the relationship between*

lean management and supply chain operations. International Journal of Entrepreneurship and Business Research, 15(3), 214–231. https://doi.org/10.1504/IJEBR.2023.132642

Leong, W. Y., Leong, Y. Z., & Leong, W. S. (2024). *Business ethics in logistics and supply chain management*. 2024 International Conference on Logistics and Industrial Engineering (ICLIE), 1–6. https://doi.org/10.1109/ICLIE61478.2024.11015398

Maheshwari, S., Gautam, P., & Jaggi, C. K. (2021). *Big Data Analytics adoption in SCM, logistics, and inventory management*. International Journal of Production Economics, 235, Article 108083. https://doi.org/10.1016/j.ijpe.2021.108083

Mirbagheri, S. (2023). *Leveraging data warehousing and decision support systems for effective supply chain management*. 2023 IEEE 8th International Conference on Smart Cloud (SmartCloud), 120–124. https://doi.org/10.1109/SmartCloud58862.2023.00028

Nofandi, F., Janaputra, M. R. B., & Pratama, R. A. (2025). *Web-based supply chain system design using blockchain*. Journal of Transport and Supply Chain Management, 19(1), Article a1136. https://doi.org/10.4102/jtscm.v19i0.1136

Pethe, S., Sahu, A., Kodarlikar, S., & Vamshidhar, M. (2024). *IoT research in supply chain management and logistics*. 2024 International Conference on Innovations and Challenges in Emerging Technologies (ICICET), 1–5. https://doi.org/10.1109/ICICET59348.2024.10616272

Reznik, N., Rudenko, S., & Pylypchuk, K. (2022). *Main characteristics of the concept of logistics and supply chain management systems*. Innovation and Sustainability, 2(3), 95–102. https://doi.org/10.31649/ins.2022.3.95.102

Smerichevska, S., Prodanova, L., & Yakushev, O. (2024). *Digitization of logistics and supply chain management.*

Intellectualization of Logistics and Supply Chain Management, 26, 113–123. https://doi.org/10.46783/smart-scm/2024-26-9

Wang, X. (2021). *Analysis of distributed nodes of logistics management information system based on supply chain*

management. 2021 5th International Conference on I-SMAC, 1–5. https://doi.org/10.1109/I-SMAC52330.2021.9640810

Yu, X., & Zhou, Q. (2020). *Design and implementation of supply chain management system based on ASP.NET*.

2020 IWCMC, 1199–1204. https://doi.org/10.1109/IWCMC48107.2020.9148185

Appendices

Appendix A

An interview was conducted on May 31, 2026, with Ms Marinic Perez, Operations Manager of the company, to gather information regarding the organization's operational processes, financial monitoring, inventory management, logistics coordination, and existing system challenges. The interview revealed that the company primarily operates as a trader of molasses, sourcing products from local sugar mills and occasionally through importation. The company's major clients include San Miguel, Tanduay, Emperador, Absolut, and other distillery companies.

The company's transaction process begins with client negotiations handled by the operations personnel. Once a deal is finalized, products are purchased from suppliers and delivered to customers. Sales invoices are then prepared and submitted to clients for payment processing. The company also manages various logistics services, including barging and tracking services, which are monitored throughout the transportation process.

A significant portion of the company's operations relies on Microsoft Excel spreadsheets. These spreadsheets are used to record purchase and selling prices, customer information, supplier details, transportation services, invoice numbers, quantities delivered and received, and profit margins. The Operations Staff consolidates and monitors all transaction data to ensure accurate tracking of expenses, revenues, and financial performance.

The interview also highlighted several operational challenges. The current system lacks centralized data management, making information accessible only to specific personnel. Updates must be manually communicated and recorded, resulting in inefficiencies and potential errors. Financial monitoring is also challenging because customer payments, supplier expenses, loans, and service costs must be manually matched and tracked. Additionally, employees rely heavily on paper-based documents, leading to duplicate printing, difficulty locating records, and inefficient document management.

A staggering amount of ₱2,343,480.55 was calculated to be the company’s operational loss due to processing errors, overdue loans, transit shrinkages and the rising price of fuel prices due to the war in Iran. Ms. Perez highlighted that these costs are still in check and are written off but stated that it would also be nicer to save up on operating expenses to be able to redirect the funds to something more worthwhile.

According to the participant, the company would benefit from a user-friendly system that centralizes operational, financial, inventory, and logistics information. The proposed system should provide real-time access to transaction records, automate data organization, simplify monitoring of contracts and deliveries, support financial tracking, and present information through dashboards and summarized reports. Such a system would improve operational efficiency, enhance data accessibility, reduce manual work, and support better decision-making within the organization

Appendix B

A stakeholder interview was conducted on September 24, 2026, with Ms. Marinic A. Perez, Operations Employee of the company, to gather feedback on the proposed website. The participant stated that the website looked good and could be a valuable tool for newly hired employees.

However, she explained that experienced employees and the company's management prefer to see the actual numbers, formulas, and calculations behind the final results and summaries. She suggested adding a transaction audit feature after every transaction to show how the results were computed.

Based on the feedback, the proposed system should include a transaction audit feature that allows users to review transaction details and verify calculations. This would help improve transparency, accountability, and user confidence in the accuracy of the system's financial records.

Appendix C

An interview was conducted on October 8, 2026, with Ms. Jewel Marie A. Perez, Operations Employee of the company, to clarify the company's purchase order, supplier sourcing, and delivery processes. The interview revealed that a single purchase order may involve multiple delivery locations, even

when the products come from the same supplier. In other cases, a single customer purchase order may be fulfilled using inventory sourced from different suppliers, depending on stock availability.

The participant explained that deliveries are completed in batches rather than all at once. For example, a customer's 10,000 metric ton purchase order may be fulfilled through several deliveries sourced from different sugar mills, such as BUSCO and CASA, until the required quantity is completed.

The interview also clarified that loan and financial matching should be handled strictly within the Finance Account to maintain proper separation between operational transactions and financial records.

Appendix D

Consultation and Testing of Deployment Conducted last Oct 20, 2026

Interviewer: How did you handle incoming Molasses Release Orders (MROs) and delivery papers before the website?

Ms. Jewel (Operations):
"It was very manual and ate up a lot of time. I used CamScanner on my phone to capture the MRO certificates to output a MRO Summary of a specific PO, but it only generated an image or PDF without reading the data. I spent around 30 to 45 minutes per batch switching back and forth between the scan and my computer, hand-typing every single detail—planter records, metric tonnes, and gross/tare weights—into Excel. When we had multiple deliveries in a day, I would easily spend 1 to 2 hours just doing manual data encoding, and a single typo meant wasting time to retrace the mistake."

Interviewer: What happened when you tried the website's document scanner (Textract) feature?

Ms. Jewel (Operations):

"It cut that time down to almost nothing. Now, I just upload the file to the website. The system reads the document and auto-fills the fields in about 5 to 10 seconds. I only spend about 3 to 5 minutes doing a quick double check before hitting save. That takes the process down from 45 minutes to under two minutes per batch, saving me at least two hours of pure typing every day."

Interviewer: How did that help you spot missing cargo or weight differences?

Ms. Jewel (Operations):

"Before, catching discrepancies could also take significant time because we had to manually calculate and cross-check MRO's spreadsheets against physical receipts. With the website automatically calculating discrepancy and price difference, we would know instantly instead of having to go and do all the rest of those things, but visual check is very important as well".

Interviewer: Overall, what was the biggest change in your day-to-day work?

Ms. Jewel (Operations):

"I can see myself saving a lot of time and the whole process feels so nicer as a young person, i am technologically inclined so when things are dealt with manually i get so irritated when i know it could be easier but i have to cater to our more experienced co workers so i have to make do with what we have. I used to spend a third of my workday just acting as a typist between CamScanner and Excel. Now that time goes directly into coordinating and verifying Schedules, MRO Summaries and others."

To summarize what ma'am Digna, ma'am Marinic and what Ma'am Diane said : they got so used to the manual process but are willing to learn how to eventually use it, they cite comfort in manually handling the finance and invoices just because it is money they are handling and they feel more in control with the process with a routine they stuck to in their years of experience. But they can see the website's potential for the incoming new hires and it's current use with ma'am jewel, ma'am Marinic also inquired about the upkeep fees.

Marinic A. Perez
Operations Manager - Consultant

Jewel Marie A. Perez
Operations Staff

Appendix E

Attached hereto are the documents submitted to us by the aforementioned employee. These documents contain the necessary context in learning the company process. The company reserved the right to redact any information it deems necessary to protect proprietary data and trade secrets.

HEINDRICH TRADING CORPORATION 2026

SI	INV. DATE	BARGE	SOURCE	PURCHASED PRICE	LOGISTIC COST TRUCKING	LOGISTICS COST BARGING	CUSTOMER	QUANTITY DELIVERED	QUANTITY RECEIVED	DISCREPANCY	SELLING PRICE	AMOUNT RECEIVABLES	
3186	1/13/2026	FRONTIER	LOPEZ	[redacted]		2,085.00	ADI	2001.63	1998.22	-3.41	[redacted]	[redacted]	
3187	1/13/2026	TRUCKING	SBTI PHILIPA		165.00		PROGREEN		1949.5			[redacted]	
3188	1/13/2026	MODESTO	CASA			2,340.00	PROGREEN	668.558	899.663	6.23		[redacted]	
			SBTI PHILIPA		165.00			224.872					
3189	1/14/2026	YANGTZE	CASA			2,340.00	PROGREEN	1124.991	1444.794	-10.95		[redacted]	
			SBTI PHILIPA		165.00			330.750					
3190	1/19/2026	TRUCKING	HTC TANK		175.00		ADI	1620.210	1627.96	7.75		[redacted]	
3206	1/19/2026	GILBERT	LOPEZ			2,085.00	ADI	1143.360	1138.68	-4.68		[redacted]	
3209	1/26/2026	NIGEL	LOPEZ			2,085.00	ADI	1929.810	1920.87	-8.94		[redacted]	
3210	1/26/2026	TRUCKING	HTC TANK		175.00		ADI	903.2	902.58	-0.62		[redacted]	
3230	2/10/2026	TRUCKING	HTC TANK		175.00		ADI	892.12	895.09	2.97		[redacted]	
3231	2/12/2026	FS2	BUSCO		165.00	2,340.00		2411.304	2367.95	-43.35		#REF!	
3234	2/25/2026	FAITH	BUSCO			1,680.00	GSMI	2207.78	2217.065	9.28		[redacted]	
	2/7/2026	FS2	BUSCO			2,340.00	HTC	2412.976	2515.94	102.96		[redacted]	
3277	3/11/2026	CHARITY	BUSCO			1,680.00	GSMI	1791.612	1760.435	-31.18		[redacted]	
3278	3/11/2026	FAITH	AABC			2,600.00	GSMI	2178.58	2167.865	-10.72		[redacted]	
3279	3/11/2026	YANGTZE	CASA		495.00	1,745.00	PROGREEN	1,478.96	1485.38	6.42		[redacted]	INV. OF CASA 1500. BL C
3280	3/11/2026	1ST KIRSTINE MARIE	CASA		495.00	2,200.00	PROGREEN	603.56				—	
							PROGREEN	120.35				—	
							PROGREEN	1,740.72	2443.48	-21.15		[redacted]	
3283	3/17/2026	CHARITY	BUSCO			2,100.00	GSMI	1742.219	1779.02	36.80		[redacted]	
3292	3/24/2026	2ND VIOLETTE	BUSCO			2,340.00	BDI	1,942.49				—	
								1,469.95	3413.323	0.89		[redacted]	
3293	3/24/2026	BARILES	CASA				PROGREEN	300				—	INV. OF CASA 1400. BL (
								1100	1331.411	0.48		[redacted]	
3295	3/27/2026	FAITH	BUSCO			2,340.00	GSMI	2205.738	2230.375	24.64		[redacted]	
3296	3/31/2026	ESPADA	BUSCO			3,500.00	BDI	1846.926	1847.164	0.24		[redacted]	
3297	3/31/2026	HARMONY	AABC			3,080.00	BDI	2500.45	2,499.92	-0.53		[redacted]	
3317	4/10/2026	2ND KIRSTEN MARIE	CASA		750.00	2,200.00	EMPERADOR	800	2439.03	-33.09		[redacted]	INV. OF CASA 2500 THE
					840.00			659.285	27.875				EDI (2439.030) + ROB 21

3318	4/10/2026	2ND KIRSTEN RAE	CASA		480.00	2,200.00	EMPERADOR	600	1755.4	25.40		370.40 MT (BY D
3354	4/10/2026				600.00			1500	370			INV. OF CASA 21
3355	4/10/2026	TRUCKING	HTC		205.00		EMPERADOR	652.8	652.31	-0.49		2,762,489.00
3357	4/15/2026	3RD VIOLETTE	BUSCO			2,680.00	BDI	3452.911	3453.22	0.31		20,850,158.80
3358	4/20/2026	HARMONY	BUSCO			2,955.76	BDI	2537.395	2536.911	-0.48		25,275,052.94
3359	4/22/2026	1ST APRIL SHEIN	BUSCO			2,680.00	BDI	2800.008	2801.383	1.38		32,727,550.87
3361	4/24/2026	HOPE	BUSCO			3,300.00	GSMI	1906.216	1917.03	10.81		20,118,815.00
3362	4/27/2026	1ST APOSTLE	CAB			2,800.00	BDI	410	2136.197	0.83		26,691,781.52
								1725.364				
3366	4/29/2026	4TH VIOLETTE	BUSCO			2,981.00	BDI	3450.16	3449.435	-0.72		56,398,262.25
3367	5/4/2026	FS2	BUSCO			3,500.00	BDI	2449.135	2450.15	1.01		40,059,750.50
3372	5/6/2026	2ND APRIL SHEIN	BUSCO			2,981.00	BDI	2796.593	2797.341	0.75		24,853,775.80
3373	5/6/2026	HOPE	BUSCO			2,800.00	BDI	1906.194	1906.771	0.58		22,004,157.24
3374	5/8/2026	AMAZON	CAB			2,940.00	EMPERADOR	1592.11	1582.69	-9.42		16,834,013.00 900 AT 3000 ANI
3384	5/13/2026	YANGTZE	CAB			2,940.00	EMPERADOR	1412.459	1424.108	11.65		20,210,333.60
3376	5/8/2026	TRUCKING	AABC ZKYARC		255.00		BDI	1556.46	1540.46	-16.00		25,186,523.00
3385	5/13/2026	3RD KIRSTEN RAE	CASA	6,591.09	925.00	2,330.00	EMPERADOR	2100.335	1745.543	2.34	11,900	20,771,861.70 1356.355 AT 856
									357.13			3,871,186.00 6591.09 IS PROR
												TRUCKING 1503
3388	5/15/2026	2ND APOSTLE	CAB			2,800.00	BDI	2112.903	2113.584	0.68		26,409,232.00
			FS2			2,906.25	HTC	2451.778	2456.891	5.11		THEN TO CBPI
			BUSCO			3,450.00	BDI	3583.73	3581.928	-1.80		58,564,522.80
3389	5/16/2026	5TH VIOLETTE	BUSCO			2,800.00	BDI	1900.97	1900.906	-0.06		30,414,496.00
3394	5/25/2026	HOPE	URC - PASSI			2,352.00	BDI	2800.11	2799.818	-0.29		44,759,295.05
3396	5/29/2026	3RD APRIL SHEIN	URC-PASSI			2,500.00	BDI	2100	2109.226	9.23		34,485,845.10
3397	5/29/2026	3RD APOSTLE	URC - SONEDCO									
			LOPEZ									
ON GOING												
		3RD KIRSTEN MARIE	CASA		850.95	2,542.000	EMPERADOR	2461.406	2460.918	-0.49	14,200	34,941,163.60 1845.415 AT 836

ON GOING												
	3RD KIRSTEN MARIE	CASA		850.95	2,542.000	EMPERADOR	2461.406	2460.918	-0.49	14,200	34,941,163.60	1845.415 AT 8360/MT A
												w/ rob from kirsten mari
	4TH KIRSTEN RAE	CASA X CAB		635.82	2,452.00	EMPERADOR	1000			14,200		BL QTY 2099.380
				635.82			1100	2100.334				835.454 AT 690/MT AND
	AMAZON	CAB			2,900.00	EMPERADOR	1561.679			14,200		PENALTY IN BRIX
	YANGTZE	CAB			2,900.00		1438.818			14,200		
	4TH KIRSTEN MARIE	CASA			2,325.00	EMPERADOR	2478.14			14,200		
	TRUCKING	HTC				EMPERADOR	499.67			14,200		
	FS2	CAB				BDI	2505.593		-2505.59	12,495		
	ESPADA	CAB			2,906.25	BDI	1865.049	1865.353	0.30	12,495		
	2ND TRUCKING	ZKYARC				BDI				16,350		
	5TH KIRSTEN RAE	LOPEZ			2,325.00	BDI	2099.68			16,000		
	6TH VIOLETTE	URC-SONEDCO			2,680.00	BDI	2329.41			16,350		TOTAL BL QTY. 2729.730
		LOPEZ					400.32					
	4TH APRIL SHEIN	URC-PASSI			2,325.00	BDI	1109.75			16,000		
		CASA				EMPERADOR	3000					

BDI CONTRACT SUMMARY AS OF 10-7-26.xlsx

File Tools Help

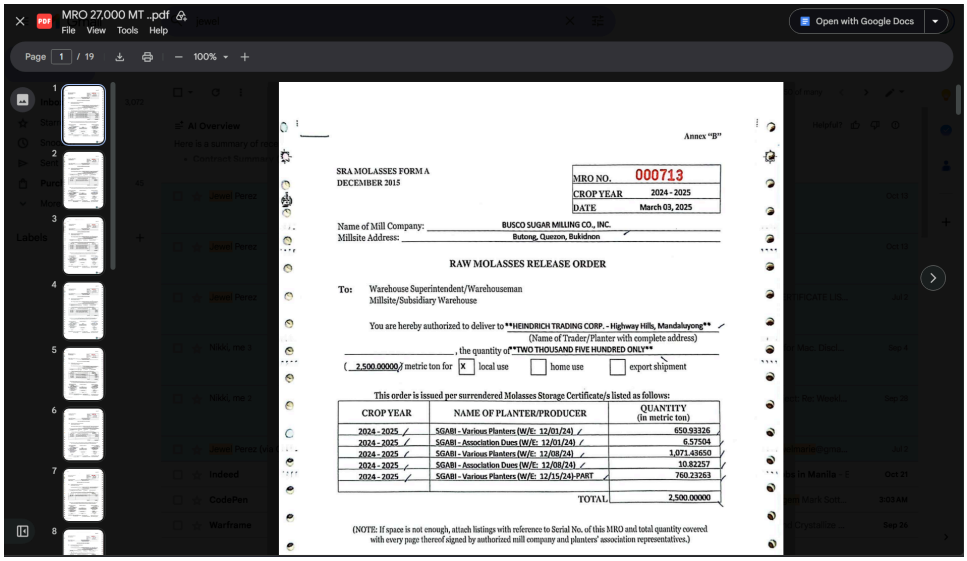
Open with Google Sheets

BDI09.18.25-MEAX	TSAI 54				
CONTRACT QTY	20,000.00	UNIT PRICE			13,395.00
DELIVERED	20,000.00				
TOTAL BALANCE	(0.00)				
BARGE	**SALES INV.**	**QTY**			
FS2	28	2405.439	**DEPOSITED**	31,248,262.93	
OCEAN LINK	29	2593.713	**DEPOSITED**	33,649,031.27	
FS2	30	2561.824	**DEPOSITED**	33,343,356.23	
MODESTO	31	939.338	**DEPOSITED**	12,239,910.74	
OCEAN LINK	32	2802.566	**DEPOSITED**	36,309,881.61	
FS2	33	2066.541	**DEPOSITED**	26,656,082.74	
BALYENA	36	3517.252	**DEPOSITED**	46,642,454.63	
BALYENA	37	3113.327	**DEPOSITED**	41,285,985.01	
TOTAL		**20,000.00**		**261,374,965.16**	

BDI11.08.25-MEAX	TSAI 54				
CONTRACT QTY	20,000.00	UNIT PRICE			13,395.00
DELIVERED	20,000.00				
TOTAL BALANCE					
BARGE	**SALES INV.**	**QTY**			
BALYENA	37	499.96	**DEPOSITED**	6,629,994.56	
ESPADA	38	1660.605	**DEPOSITED**	22,021,365.94	
ESPADA	39	1984.157	**DEPOSITED**	26,312,005.18	
FS2	41	2400.028	**DEPOSITED**	31,826,891.31	
ESPADA	43	1883.787	**DEPOSITED**	24,980,993.60	
FAITH	44	2220.087	**DEPOSITED**	24,980,993.60	
FS2	47	2301.837	**DEPOSITED**	30,524,775.55	
FAITH	48	2209.988	**DEPOSITED**	29,306,761.37	
HOPE	49	1900.154	**DEPOSITED**	25,199,509.18	
ESPADA	50	1866.543	**DEPOSITED**	24,753,513.54	
BARILES	51	1072.85	**DEPOSITED**	17,364,522.79	
TOTAL		20,000.00		**263,901,326.62**	

BDI08.07.25-HTC	TSAI 54				
CONTRACT QTY	2,500.00	UNIT PRICE	12,800.00		
DELIVERED	1,744.46				
TOTAL BALANCE	755.54				
BARGE	**SALES INV.**	**QTY**			
AMAZON	2948	159.352	**DEPOSITED**	2,019,308.54	
MODESTO	3009	594.157	**DEPOSITED**	7,513,665.41	

BDI09.25.25-HTC	TSAI 51-52				
CONTRACT QTY	10,000.00	UNIT PRICE	12,495.00		
DELIVERED	10,002.93				
TOTAL BALANCE	(2.93)				
BARGE	**SALES INV.**	**QTY**			
PHIL CEBU	3005	1091.201	**DEPOSITED**	13,498,210.93	
1ST APOSTLE	3362	2136.197	**DEPOSITED**	26,424,863.70	



MRO 27,000 MT ..pdf

Annex "B"

SRA MOLASSES FORM A
DECEMBER 2015

MRO NO.	000713
CROP YEAR	2024 - 2025
DATE	March 03, 2025

Name of Mill Company: BUSCO SUGAR MILLING CO., INC.
Millsite Address: Butong, Quezon, Bukidnon

RAW MOLASSES RELEASE ORDER

To: Warehouse Superintendent/Warehouseman
Millsite/Subsidiary Warehouse

You are hereby authorized to deliver to **HEINDRICH TRADING CORP. - Highway Hills, Mandaluyong**
(Name of Trader/Planter with complete address)
, the quantity of **TWO THOUSAND FIVE HUNDRED ONLY**
(2,500.00000) metric ton for [X] local use [ ] home use [ ] export shipment

This order is issued per surrendered Molasses Storage Certificate/s listed as follows:

CROP YEAR	NAME OF PLANTER/PRODUCER	QUANTITY (in metric ton)
2024 - 2025	SGABI - Various Planters (W/E: 12/01/24)	650.93326
2024 - 2025	SGABI - Association Dues (W/E: 12/01/24)	6.57504
2024 - 2025	SGABI - Various Planters (W/E: 12/08/24)	1,071.43650
2024 - 2025	SGABI - Association Dues (W/E: 12/08/24)	10.82257
2024 - 2025	SGABI - Various Planters (W/E: 12/15/24)-PART	760.23263
	TOTAL	2,500.00000

(NOTE: If space is not enough, attach listings with reference to Serial No. of this MRO and total quantity covered with every page thereof signed by authorized mill company and planters' association representatives.)

ABSOLUT DISTILLERS, INC.
Bo. Malaruhatan, Lian, Batangas 4216 Philippines
VAT REG. T.I.N.: 000-617-524-00000
Tel. No.: (02) 8245-2613

PURCHASE ORDER

SUPPLIER CODE : VH018
SUPPLIER NAME : HEINDRICH TRADING CORP
ADDRESS : UNIT 305 BMRC BANAHAW HIGHWAY HILLS MANDALUYONG
T.I.N. : 000-121-227-00000

DOCUMENT NO. : 121005566
DOCUMENT DATE : July 09, 2026
PAYMENT TERMS : AS STATED

ITEMCODE	DESCRIPTION	QUANTITY	UNIT	UNIT PRICE	TOTAL AMOUNT
PR # 111003605					
501011000	MOLASSES	10,000.000	mt	14,500.00	145,000,000.00
	Total Amount		:		**145,000,000.00**

PR# FMD-2026-0001-ADB
Crop Year: 2025-2026
pH: 5.0 – 5.70
% U.S.: Not higher than 4.5%
% SA: Not higher than 8%
% VA: Not higher than 8,000 ppm
Tsai: Min 55%
Brix: Min 80%
Tolerance: ±10% At Seller Option
Payment: 30 Days after Delivery to Absolut Distiller Inc.
Time of Delivery: August 2026
Source: Balayan Sugar Mill;
MRO shall be sent upon withdrawal; Delivered to Absolut and the unit price included freight cost
* Note *
1. Tsai and brix to be tested is the average of every five (5) trucks but in no way should it delay the discharging of the molasses from the barges vessels.
2. For deficiency below 55%, seller will adjust price in the commercial invoice at 1/55 of the contract price per percent total sugar content below 55%,.
3. In case there is a difference of more than 1% between the analysis result by SGS during loading compared with what was found by Absolut Distillers, Inc. during the discharged at their plant, then another composite sample drawn from the discharged at Absolut Distillers, Inc. shall be sent to the Sugar Regulatory Administration laboratory for free analysis. The result of the referee sample from the Sugar Regulatory Administration, Quezon City will be taken as final.

Appendix F

Pictures of our Site Visit Conducted last October 20th 2026

JAMES BYRD
ADI Gallery - San Francisco

Appendix G

Scope of Work (SOW), Functional Specifications, Iterations Approval

Scope Category	Boundary	Technical & Operational Description
Procurement & Contracts	In-Scope	Centralized digital logging of upstream supplier agreements (BUSCO) and downstream customer purchase orders (San Miguel, Tanduay, Emperador, ADI, GSMI) with status visualizers.
Logistics Ledger	In-Scope	Tracking land/sea transport loads, freight costs, barge logs, and tracking 50% upfront logistics deposits.
Volume Variance Tracking	In-Scope	Pure-software Celery/Redis background engine to calculate transit weight discrepancies against a 1% physical shrinkage tolerance threshold, firing visual alerts upon breach.
Financial Lifecycle Engine	In-Scope	Dynamic bank loan interest accrual engine, loan aging countdown clocks, physical cheque logs, and payment-to-expense matching linking Sales Invoices to

Scope Category	Boundary	Technical & Operational Description
		internal Cash Vouchers.
Access & Audit Control	In-Scope	Role-based permissions (Admin vs. Standard Employee) with an unalterable, timestamped database audit log for cheque movements and financial state changes.
Cloud Architecture	In-Scope	Cloud-hosted web system (Django, PostgreSQL, NGINX, Gunicorn) accessible via standard desktop web browsers without local hardware server dependencies.
Hardware Integration	Out-of-Scope	Excludes physical hardware arrays, RFID tags, barcode scanners, or physical barge GPS units.
Accounting & Tax	Out-of-Scope	Excludes general ledgers, macro balance sheets, multi-currency ledgers, tax localization modules, and automated tax filing.
Macro Financial	Out-of-Sco	Excludes overall corporate P&L statement generation to

Scope Category	Boundary	Technical & Operational Description
Auditing	pe	maintain strict company financial confidentiality.
Public E-Commerce	Out-of-Sco pe	Excludes customer-facing storefronts, broker portals, or online payment gateways.
Regulatory Automation	Out-of-Sco pe	Excludes automated enforcement of Sugar Regulatory Administration (SRA) rules (e.g., 3:1 local-to-import allocation constraints).

Iteration 1 Incremental Multi-Batch Delivery Ledger & Relational Schema Overhaul

Stakeholder Input: Commodity deliveries are never completed in a single shipment. A single PO (e.g., 10,000 MT) for a client like ADI is fulfilled incrementally over time. The initial 2,000 MT may be sourced from BUSCO and delivered directly to ADI; once BUSCO stock exhausts, subsequent batches are sourced from CASA until the full 10,000 MT contract is satisfied.

Implementation:

Refactored database tables from a single flat record into a PO Header (Parent) and 1-to-Many Delivery Ledger (Child) relationship.

Dynamic Remaining MT Formula: remaining_mt = total_contracted_mt - SUM(delivered_mt).

Automated Status Transition: When remaining_mt equals 0, po_status automatically toggles to 'Completed'.

Validation Engine: Implemented API and ORM guards preventing delivery records where delivered_mt > remaining_mt (HTTP 400 rejection).

Iteration 2 Transaction Audit Summaries & Senior Stakeholder Usability

Stakeholder Input (Ms. Marinic A. Perez): Executive management and senior staff require explicit visibility into underlying calculations, mathematical formulas, and step-by-step transaction breakdowns following delivery completions.

Implementation:

Integrated audit summary panels displaying explicit formulas (e.g., Gross Margin = Selling Price - Sourcing Cost - Freight Rate; Shrinkage % = (Loaded - Received) / Loaded * 100).

Implemented High-Legibility Elderly Usability Mode (A+) expanding body typography to 17px base, heading levels to 24-32px, and interactive elements to 48px touch targets.

Streamlined terminology and decluttered modal layouts across operational screens.

Iteration 3 System Integrity, Timezone Synchronization & Document AI

Stakeholder Input : Processing MROs from scanning to typing consumes a lot of time throughout a business day, is it possible to streamline that process using your website?

Implementation:

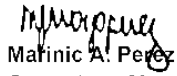
Asia/Manila (PHT) Timezone Consolidation: Standardized database timestamps, audit logs, and transaction creation times strictly to Philippine Standard Time.

UI Layout Symmetry: Positioned the immutable activity audit log (powered by django-simple-history) directly below the financial matching tab to establish visual balance.

Document AI OCR (MRO Permitting): Integrated AWS Textract parsing background workers. Users can manually input MRO data or upload PDF/images to auto-fill metadata with extraction confidence indicators and live preview modals.

Jewel Marie A. Perez

Operations Staff



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